Math Notes

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Pre 2023

1.1 Mean and Variance, and the arbitrariness thereof

For some time I have wondered about the arbitrariness around the mean and variance. For example why the arithmetic mean instead of the geometric or root-mean-squared? And why square root the variance to give the standard deviation?

Well the strictness of Markov's and Chebyshev's might provide a reason. Both rely of the conditional expected value, so just to reiterate:

$$E[X|X \geq a] \geq a$$

Since everything X can be is greater then a it's expected value must be greater than a. Notice the strictness of the inequality, this will be used to make the following inequalities much stricter.

Markov

$$\begin{aligned}
\mu =& E[X] \\
=& P(X \leq a)E[X|X \leq a] + P(X \geq a)E[X|X \geq a] \\
\geq& 0 \cdot E[X|X \leq a] + P(X \geq a)a \\
\frac{\mu}{a} \geq& P(X \geq a)
\end{aligned}$$

Chebyshev

$$\begin{aligned}
& E[(X-a)^2] \\
=& P(|X-a| \leq b)E[(X-a)^2||X-a| \leq b] + P(|X-a| > b)E[(X-a)^2||X-a| > b]
\end{aligned}$$

A General Relation

Assume:

$$f(S') \geq 0, \quad g(S) \geq 0$$

Then through:

$$E[f(X)] = P(X \in S)E[f(X)|X \in S] + P(X \in S')E[f(X)|X \in S']$$

We have:

$$1 - \frac{E[g(X)]}{E[g(X)|X \in S']} \leq P[X \in S] \leq \frac{E[f(X)]}{E[f(X)|X \in S]}$$

With dual equality if:

$$f = 1_S, \quad g = 1_{S'}$$

Covariance

Lets try to find the lest squares regression between X and Y such that:

$$E[X] = E[Y] = 0, E[X^2] = E[Y^2] = 1$$

Since the expected values are both zero the line is through the origin

$$\begin{aligned}\sum_n (mx_n + c - y_n)^2 =& nE[(mX + c - Y)^2] \\ =& n\Big(E[m^2X^2] + E[c^2] + E[Y^2] + E[2cmX] + E[-2mXY] + E[-2cY]\Big) \\ =& n(m^2 + c^2 + 1 - 2mE[XY])\end{aligned}$$

Trying to minimize this value by our selection of trivially gets:

$$c = 0, \quad m = E[XY]$$

Just expanding the definitions gives:

$$COV[X, Y] = E[(X - E[X])(Y - E[Y])] = E[XY] = m$$

Hence the covariance can 'naturally' be interpreted and the first order function between the valuables.

$$E[f(X)] \approx E[f_0 + f_1X + f_2X^2/2] = f_0 + \mu f_1 + \sigma^2 f_2/2$$

$$E[f(X)] \approx E[f(\mu) + (X - \mu)f'(\mu) + (X - \mu)^2/2f''(\mu)] = f(\mu) + \frac{f''(\mu)}{2}\sigma^2$$

1.2 Discontinuities in a Non-decreasing Function

Let f be a non-decreasing function.

Define the jump function J as:

$$J(x) = \inf\{f(t)|t > x\} - \sup\{f(t)|t < x\}$$

This function is well defined since the sets are appropriately bound by $f(x)$. And it is clear that $J(d) \neq 0$ iff d is a discontinuity and that J is non-negative.

Let $U = (x_0, x_1)$. For $d_n \in U$ with $n < m \Rightarrow d_n < d_m$ we have:

$$f(x_1) - f(x_0) \geq \sum_k J(d_k)$$

Proof:

$$\begin{aligned}
&\sum_k J(d_k) \\
=&\inf\{f(t)|t > d_n\} - \sup\{f(t)|t < d_0\} + \sum_k \left[\inf\{f(t)|t > d_{k-1}\} - \sup\{f(t)|t < d_k\}\right] \\
\leq& f(d_n) - f(d_0) + \sum_k \left[f(d_{k-1}) - f(d_k)\right] \\
\leq& f(d_n) - f(d_0) \\
\leq& f(x_1) - f(x_0)
\end{aligned}$$

Let $S_n = \{d \in U|J(d) > \frac{1}{n}(f(x_1) - f(x_0))\}$ From the previous inequality there are at most n elements in S_n. Hence:

$$\{d \in U|J(d) > 0\} = \bigcup_n \left\{d \in U|J(d) > \frac{1}{n}(f(x_1) - f(x_0))\right\}$$

Is countable, hence the number of discontinuities of f on U is countable.

By corollary the discontinuities of a non-decreasing function on $\mathbb{R}$ are countable:

Let f be a non-decreasing function on $\mathbb{R}$. Let $X_n = (n-1, n+1)$, clearly $\mathbb{R} = \bigcup_n X_n$[1]. Assume f has an uncountable number of discontinuities then at least one X_n contains uncountable discontinuities. Otherwise there would be a countable set of countable sets of discontinuities, making them countable. But f being non-decreasing function and having an uncountable number of discontinuities in X_n is a contradiction.

[1]Having them overlap simplify the proof by avoiding literal edge-cases.

1.3 Integrals and Symmetry

1.3.1 Domain Symmetry

Let U be a subset of $\mathbb{R}^n$ and let $\phi : U \to U$ be a function such that:

$$\begin{aligned} \phi(U) &= U \\ |\det \phi'(\mathbf{u})| &= 1 \end{aligned}$$

Basically ϕ is a linear permutation[2] on U, this is a symmetry in the most direct sense. We obtain the following:

$$\begin{aligned} \int_U f(\mathbf{v})\, d\mathbf{v} &= \int_{\phi(U)} f(\mathbf{v})\, d\mathbf{v} \\ &= \int_U f(\phi(\mathbf{u}))|\det \phi'(\mathbf{u})|\, d\mathbf{u} \\ &= \int_U f(\phi(\mathbf{u}))\, d\mathbf{u} \end{aligned}$$

In particular we get:

$$0 = \int_U \big(f(\mathbf{u}) - f(\phi(\mathbf{u}))\big)\, d\mathbf{u}$$

This integral is important since a function can be split into a vanishing and non-vanishing part:

$$\begin{aligned} f(\mathbf{u}) =& \frac{1}{2}\big(f(\mathbf{u}) + f(\phi(\mathbf{u})) + \frac{1}{2}\big(f(\mathbf{u}) - f(\phi(\mathbf{u}))\big) \\ \int_U f(\mathbf{u})\, d\mathbf{u} =& \frac{1}{2}\int_U \big(f(\mathbf{u}) + f(\phi(\mathbf{u}))\, d\mathbf{u} + \frac{1}{2}\int_U \big(f(\mathbf{u}) - f(\phi(\mathbf{u}))\big)\, d\mathbf{u} \\ =& \frac{1}{2}\int_U \big(f(\mathbf{u}) + f(\phi(\mathbf{u}))\, d\mathbf{u} \end{aligned}$$

[2]Note that ϕ being a permutation requires that if the magnitude of the determinate is constant it must be unity, this can be seen by setting f to a constant.

For example, consider the classic odd function on an integral centered at 0.

$$\phi(x) = -x$$
$$U = [-1, 1]$$

We get the familiar:

$$\int_{-1}^{1} f(x)\,dx = \frac{1}{2}\int_{-1}^{1} \big(f(x) + f(-x)\big)\,dx$$

The utility of this relation can be seen by applying it to the basis of a class of function. Let $V = \langle 1, x, x^2 \rangle$, this is a basis for all parabolas. Notice that x base element vanishes, simplify the evaluation of integrals.

For a 2-D example recall that for two dimensional change of variables:

$$(x, y) = \phi(u, v)$$

We have:

$$|\det \phi'(\mathbf{v})| = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

The rotation symmetry for a regular triangle is:

$$(x, y) = \frac{1}{2}(-u - \sqrt{3}v, \sqrt{3}u - v)$$

Obviously the symmetries act like a group and with functions being a vector space we can use group representations.

1.3.2 Function Symmetry

Let f and ϕ be functions such that:

$$f(t) = \phi'(t) f(\phi(t))$$

Then for arbitrary x_0 and x_1 we have:

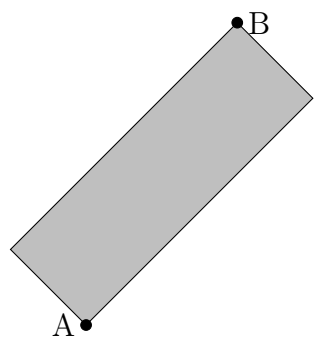
$$\begin{aligned}
\int_{x_0}^{x_1} f(t)\,dt &= \int_{\phi(x_0)}^{\phi(x_1)} \phi'(t) f(\phi(t))\,dt \\
&= \int_{\phi(x_0)}^{\phi(x_1)} f(t)\,dt \\
&= \int_{x_1}^{\phi(x_1)} f(t)\,dt + \int_{\phi(x_0)}^{x_1} f(t)\,dt \\
\int_{x_0}^{\phi(x_0)} f(t)\,dt &= \int_{x_1}^{\phi(x_1)} f(t)\,dt
\end{aligned}$$

Hence the integral value is independent of x_n, in particular if ϕ has a fixed point then the integral is zero.

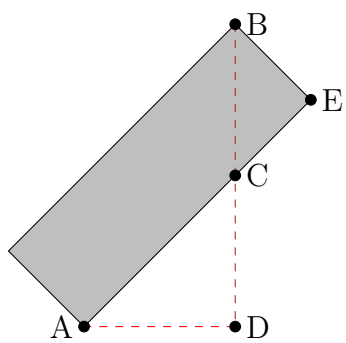
1.4 Causal Metric

1.4.1 Basic Geometry

Consider two points A and B that we wish to find the area of the rectangle between them in terms of their coordinates on an axis at a 45° angle.



Drop an altitude from B to line up horizontally with A and label the end-point D. The length |AD| and |BD| are the coordinates.



The triangles △ACD and △BCE are $45° - 90° - 45°$ triangles meaning:

$$\begin{aligned}
|\mathrm{AC}| =& \sqrt{2}|\mathrm{AD}| \\
|\mathrm{BE}| =& \frac{1}{\sqrt{2}}|\mathrm{BC}| \\
=& \frac{1}{\sqrt{2}}(|\mathrm{BD}| - |\mathrm{CD}|) \\
=& \frac{1}{\sqrt{2}}(|\mathrm{BD}| - |\mathrm{AD}|)
\end{aligned}$$

Hence the (signed) area of the rectangle is:

$$\begin{aligned}|\mathrm{BE}| \cdot |\mathrm{AE}| =& |\mathrm{BE}| \cdot (|\mathrm{AC}| + |\mathrm{CE}|) \\ =& (|\mathrm{BD}| - |\mathrm{AD}|)(|\mathrm{BD}| + |\mathrm{AD}|) \\ =& |\mathrm{BD}|^2 - |\mathrm{AD}|^2\end{aligned}$$

1.4.2 Causal Metric

This construction supplies some intuition for Minkowski Metric. Since if we interpret the plane as the set of events where the horizontal component is space-like and the vertical is time-like. The reason the rectangle is at a 45° is because that's the maximum speed of propagation. The rectangle's area is a measure of the amount of events in-between the two. And the sign of the area is the type of causal connection. Wether the points are in the way (space-like), or another events are a means by which the earlier effect the later (time-like).

1.5 Wave Equation

1.5.1 Symmetry

Given some function $u(\vec{x}, t)$ on space and time we want to understand it's dynamics under the following assumptions:

1. **Reversible**, the dynamics should be symmetric in respect to reversing time.
2. **Isotropic**, the function should be symmetric in respect to all directions.
3. **Relative**, the absolute value doesn't matter, only changes.
4. **Perturbation**, the changes should be small.

The wave equation is a natural conclusion from from these constraints:

$$\frac{\partial^2}{\partial t^2} u = c \sum_i \frac{\partial^2}{\partial x_i^2} u$$

Because the change change is small we start by considering a Taylor expansion and try to get the lowest order terms. Because it's relative we ignore the 0th order terms. Because it's reversible we ignore the 1st order terms Giving:

$$\frac{\partial^2}{\partial t^2} u = \sum_{i,j} w_{i,j} \frac{\partial^2}{\partial x_i \partial x_j} u$$

Because it's isotropic all the $\frac{\partial^2}{\partial x_i^2}$ coefficient need to be equal. And without loss of generality we can assume a basis where the cross terms vanish, giving:

$$\frac{\partial^2}{\partial t^2} u = \sum_i cu \frac{\partial^2}{\partial x_i^2} = c \sum_i \frac{\partial^2}{\partial x_i^2} u \quad \square$$

1.5.2 Old Symmetry Attempt

(Not sure what I was doing here, lol. Seems like I went on the wrong path by not including t) Lets start with with a function of space $u(\vec{x})$. Lets assume it is continuous such that:

$$u(\vec{x} + \vec{r}) = u(\vec{x}) + \sum_i r_i \frac{\partial}{\partial x_i} u(\vec{x}) + \frac{1}{2} \sum_{i,j} r_i r_j \frac{\partial^2}{\partial x_i \partial x_j} u(\vec{x})$$

Lets remove all terms that aren't reversible:

$$u(\vec{x}+\vec{r}) = u(\vec{x}) + \frac{1}{2}\sum_i r_i^2 \frac{\partial^2}{\partial x_i^2} u(\vec{x})$$

Lets impose localization:

$$c^2 r_0^2 = \sum_{i>0} r_i^2$$

Not sure exactly how this part works but we get the sign we need one example is to only keep terms that have $c^2 r_0^2 = r_i^2$

1.5.3 1-D

I need to remind myself how to solve the 1-D case as a stepping stone for something else. Reminder that the 1-D form, with unitary speed, is:

$$\frac{\partial^2}{\partial t^2} u = \frac{\partial^2}{\partial x^2} u$$

Observe that plane waves where angular frequency and wave number have the same magnitude solve this:

$$\exp(ik(x \pm t))$$

This suggests the use of Fourier transform, where the $\pm$ is used to fit the solution to initial value and derivative:

$$u = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (a_+(k) e^{ikt} + a_-(k) e^{-ikt}) e^{ikx}\, dk$$

Assume $f(x) = u(x,0)$ and $h(x) = \left[\frac{\partial}{\partial t} u\right](x,0)$. Equating Fourier transforms gives:

$$\begin{aligned} \hat{f}(k) =& a_+(k) + a_-(k) \\ \hat{h}(k) =& ik(a_+(k) - a_-(k)) \end{aligned}$$

Hence:

$$u = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (\hat{f}(k)\cos(kt) + \hat{h}(k) k^{-1} \sin(kt)) e^{ikx}\, dk$$

This can be processed further through the convolutions theorem. (Because of the table of transforms I have it will be easier sin is turned into normalized sinc and rect cuts off at $\frac{1}{2}$):

$$u = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \left(\hat{f}(k)\cos(kt) + \hat{h}(k) t \operatorname{sinc}\left(\frac{kt}{\pi}\right) \right) e^{ikx}\, dk$$

$$\begin{aligned}
u =& \left[f(\tau) * \frac{1}{2}(\delta(\tau - t) + \delta(\tau + t)) \right](x) + \frac{\operatorname{sgn}(t)}{2} \left[h(\tau) * \operatorname{rect}\left(\frac{\tau}{2t}\right) \right](x) \\
=& \frac{1}{2}(f(x-t) + f(x+t)) + \frac{\operatorname{sgn}(t)}{2} \left[h(\tau) * \operatorname{rect}\left(\frac{\tau}{2t}\right) \right](x) \\
=& \frac{1}{2}(f(x-t) + f(x+t)) + \frac{\operatorname{sgn}(t)}{2} \int_{x-t}^{x+t} h(\tau)\, d\tau
\end{aligned}$$

Which is such a cool equation! It shows the reversibility and isotropic nature of the solution but the evenness of x and t. It shows the limit on how fast information travels by the convolution with rect.

I don't really like the sgn function there and I'm not convinced it isn't some kind of error. So ignoring it for the following bit (it's mostly ± 1 anyway), see how we can recover the conditions on f and h:

$$\begin{aligned}
u(x,0) =& \frac{1}{2}(f(x-0) + f(x+0)) + \frac{1}{2} \int_{x-0}^{x+0} h(\tau)\, d\tau \\
=& f(x) \\
\left[\frac{\partial}{\partial t} u \right](x,t) =& \frac{1}{2} - (f(x-t) + f(x+t)) + \frac{1}{2}(h(x-t) + h(x+t)) \\
\left[\frac{\partial}{\partial t} u \right](x,0) =& h(x)
\end{aligned}$$

It almost seems obvious! Like I should have gone straight to this and not bothered with the Fourier stuff. At least I learn something, I guess.

1.6 Unit Fraction

Theorem: Consider a function $f : X \to \mathbb{R}$ where $\mathrm{cl}(\mathrm{Im}(f))$ is bounded and countable. Then for any $n \in \mathbb{N}$ and interval $U \subset \mathbb{R}$ we have $n\,\mathrm{Im}(f) \not\supseteq \mathbb{Q} \cap U$. [3]

Proof: If we assume that $n\,\mathrm{Im}(f) \supseteq \mathbb{Q} \cap U$ we get:

$$|\,\mathrm{cl}(n\,\mathrm{Im}(f))| \geq |\,\mathrm{cl}(\mathbb{Q} \cap U)| = |U| = \aleph_1$$

However by the compactness of $\mathrm{cl}(\mathrm{Im}(f))$:

$$\mathrm{cl}(n\,\mathrm{Im}(f)) = n\,\mathrm{cl}(\mathrm{Im}(f))$$

Hence by the countability of $\mathrm{cl}(\mathrm{Im}(f))$:

$$|\,\mathrm{cl}(n\,\mathrm{Im}(f))| = |n\,\mathrm{cl}(\mathrm{Im}(f))| \leq |\,\mathrm{cl}(\mathrm{Im}(f))|^n = \aleph_0^n = \aleph_0$$

This contradicts $|\,\mathrm{cl}(n\,\mathrm{Im}(f))| \geq \aleph_1$ and hence $n\,\mathrm{Im}(f) \not\supseteq \mathbb{Q} \cap U$.

Corollary: The function $f : \mathbb{N}_{>0} \to \mathbb{R}$ where $f(x) = 1/x$ meets the function requirements. Hence for every interval there is a rational number that can't be represented with n unit fractions.

I want to try a more aesthetically pleasing formulation by separating out the set theory from the topology from the $\mathbb{Q}$ specifics.

Theorem: Let X and Y be sets. Then $|Y| > |X|^n$ implies $nX \not\supseteq Y$.
Proof: By contradiction on set size.

Theorem: If X is countable and Y is non-countable then $nX \not\supseteq Y$ for all $n \in \mathbb{N}_{>0}$.
Proof: The previous theorem using $\aleph_1 > \aleph_0^n$ for all $n \in \mathbb{N}_{>0}$

Theorem: If X is compact, $\mathrm{cl}(X)$ countable, and Y non-countable then $\mathrm{cl}(nX) \not\supseteq Y$ for all $n \in \mathbb{N}_{>0}$.
Proof: The previous theorem using $\mathrm{cl}(nX) = n\,\mathrm{cl}(X)$ from compactness.

Now just rework the corollary a bit for it to fit here.

[3] In this section nS means $\{\sum_i s_i | s \in S^n\}$ and not $\{ns | s \in S\}$

1.7 Winquist's identity

The following form of the Winquist's identity was on an old hard drive:

$$\begin{aligned}
&\prod_{n\geq 1}(1-ax^{n-1})(1-a^{-1}x^n)(1-bx^{n-1})(1-b^{-1}x^n)\\
&\quad\times(1-ab^{-1}x^{n-1})(1-abx^{n-1})(1-a^{-1}b^{-1}x^n)(1-x^n)^2\\
&=\sum_{i\in\mathbb{N}_{\geq 0}}\sum_{j\in\mathbb{Z}}(-1)^{i+j}(b^{-3j}-b^{3j+1})(a^{-3i}-a^{3i+3})\\
&\quad\times(b^{-3i+2}-b^{3i-1})(a^{-3j+1}-a^{3j+2})x^{\frac{j(3j+1)}{2}+\frac{3i(i+1)}{2}}
\end{aligned}$$

A quick Google make the identity verifies that the identity looks right. With the exception of the ranges of the RHS summation, maybe i and j should be switched.

Either way this is a cool identity and captures something interesting. And that would definitely be helpful in generating functions. In particular reciprocating the relation and multiplying by the RHS give a weighted partition that has a relatively sparse recursive relation.

1.8 Rational Tangent

Consider the following relation:

$$\begin{aligned}\cos(n\phi) + i\sin(n\phi) =& (\cos(\phi) + i\sin(\phi))^n \\ =& (\cos(\phi)(1 + i\tan(\phi)))^n \\ =& \cos(\phi)^n(1 + i\tan(\phi))^n\end{aligned}$$

The product of two complex numbers with rational real and imaginary part is a complex number with rational real and imaginary part.

To see this observe that we only use multiplication, addition, and subtraction to get the real and imaginary components of the product from the component of the factors and that these operations between from rational numbers produce rational numbers.

Hence, if $\tan(\phi)$ is rational the there exists some rational numbers p, q such that:

$$(1 + i\tan(\phi))^n = p + qi$$

Substituting this into the original formula:

$$\cos(n\phi) + i\sin(n\phi) = \cos(\phi)^n(p + qi)$$

And equating real and imaginary components gives:

$$\tan(n\phi) = \frac{\sin(n\phi)}{\cos(n\phi)} = \frac{\cos(\phi)^n p}{\cos(\phi)^n q} = \frac{p}{q}$$

Hence $\tan(\phi)$ being rational implies $\tan(n\phi)$ is as well.

I think this result was meant to be part of a larger argument, but I have forgotten what the larger one is. One point that I think will be relevant is using similar arguments with:

$$\frac{1}{\cos(\phi) + i\sin(\phi)} = \cos(\phi) - i\sin(\phi)$$

1.8.1 Brute Force

Another proof of the same result, done by brute force, was in the same notes on my hard-drive. Presumably done as a sanity check before figuring out the

better method:

$\tan(\phi)$ being rational is the same as saying that $r\sin(\phi) = \cos(\phi)$ for some rational number r.

$$\begin{aligned}
\sin(n\phi) + i\cos(n\phi) &= \exp(in\phi) \\
&= \exp(i\phi)^n \\
&= (\sin(\phi) + i\cos(\phi))^n \\
&= \sum_{k=0}^{n} \binom{n}{k} i^k \cos(\phi)^k \sin(\phi)^{n-k} \\
&= \sum_{k=0}^{n} \binom{n}{k} i^k r^k \sin(\phi)^n \\
&= \sin(\phi)^n \left(\sum_{k=0}^{2k\leq n} \binom{n}{2k} (-1)^k r^{2k} + i \sum_{k=0}^{2k+1\leq n} \binom{n}{2m+1} (-1)^k r^{2k+1} \right)
\end{aligned}$$

Hence:

$$\tan(n\phi) = \frac{\sum_{k=0}^{2k\leq n} \binom{n}{2k} (-1)^k r^{2k}}{\sum_{k=0}^{2k+1\leq n} \binom{n}{2k+1} (-1)^k r^{2k+1}}$$

1.9 Old Geometry

Bellow are are collection of proofs from high school that I saved under "geometry". It includes triangle groups and two (half complete) geodesics.

1.9.1 Triangle Groups

Are a way of understanding the rotations of platonic solids. Fill in latter

Subgroup chain

Let:

$$\iota^2 = \lambda^4 = \kappa^3 = 1$$

With and those be the lowest powers that do so. Additionally have:

$$\lambda = \iota\kappa$$

This is a cube, ι rotates around a edge, κ rotates around a vertex, λ rotates around a face.

Let:

$$i = \lambda^2,\ k = \lambda\iota,\ k = \kappa^2$$

We have:

$$\begin{aligned} i^2 =& \lambda^4 \\ =& 1 \\ k^3 =& \kappa^6 \\ =& 1 \\ l^3 =& (\lambda\iota)^3 \\ =& (\iota\kappa\iota)^3 \\ =& \iota\kappa^3\iota \\ =& 1 \end{aligned}$$

These are the lowest powers to do so. Minimality is trivial for i and k by the

minimality of κ and ι. For l assume $l^2 = 1$ then:

$$\begin{aligned} l^2 =& 1 \\ \lambda\iota\lambda\iota =& 1 \\ \iota\lambda\iota =& \lambda^3 \\ \kappa\iota =& \lambda^3 \\ \kappa\iota\lambda =& \lambda^4 \\ =& 1 \\ \kappa^2 =& 1 \end{aligned}$$

We have the relation:

$$\begin{aligned} ik =& \lambda^3\iota \\ =& \lambda^3\iota\kappa^3 \\ =& \lambda^3\lambda\kappa^2 \\ =& \kappa^2 \end{aligned}$$

Hence the rotations of a regular tetrahedron are a subgroup of the rotations of a cube.

1.9.2 Torus

Let a torus be parametrized by:

$$\begin{aligned} r =& (x, y, z) \\ x =& (R + r\cos(\theta))\cos(\phi) \\ y =& (R + r\cos(\theta))\sin(\phi) \\ z =& r\sin(\theta) \end{aligned}$$

The partials of r are given by:

$$\begin{aligned} \frac{\partial r}{\partial \phi} =& (-(R + r\cos(\theta))\sin(\phi), (R + r\cos(\theta))\cos(\phi), 0) \\ \frac{\partial r}{\partial \theta} =& (-r\sin(\theta)\cos(\phi), -r\sin(\theta)\sin(\phi), r\cos(\theta)) \\ \frac{\partial r}{\partial \phi} \cdot \frac{\partial r}{\partial \phi} =& (R + r\cos(\theta))^2 \\ \frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \theta} =& r^2 \\ \frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \phi} =& 0 \end{aligned}$$

Use the partial to get the line element:

$$\begin{aligned}
\dot{r}^2 &= \left(\frac{\partial r}{\partial \phi}\dot{\phi} + \frac{\partial r}{\partial \theta}\dot{\theta}\right)^2 \\
&= \frac{\partial r}{\partial \phi} \cdot \frac{\partial r}{\partial \phi}\dot{\phi}^2 + \frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \theta}\dot{\theta}^2 + 2\frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \phi}\dot{\phi}\dot{\theta} \\
&= (R + r\cos(\theta))^2\dot{\phi}^2 + r^2\dot{\theta}^2
\end{aligned}$$

Treating the line element as the integrand in Lagrangian integral:

$$L = \frac{1}{2}\dot{r}^2$$

$$\begin{aligned}
\frac{\partial L}{\partial \phi} =& \frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} \\
0 =& \frac{d}{dt}(R + r\cos(\theta))^2\dot{\phi} \\
=& \left[\dot{\theta}\frac{\partial}{\partial \theta} + \ddot{\phi}\frac{\partial}{\partial \dot{\phi}}\right](R + r\cos(\theta))^2\dot{\phi} \\
=& (R + r\cos(\theta))^2\ddot{\phi} - 2r\sin(\theta)(R + r\cos(\theta))\dot{\phi}\dot{\theta} \\
\ddot{\phi} =& \frac{2r\sin(\theta)}{R + r\cos(\theta)}\dot{\phi}\dot{\theta}
\end{aligned}$$

And again:

$$\begin{aligned}
\frac{\partial L}{\partial \theta} =& \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} \\
-r\sin(\theta)(R + r\cos(\theta))\ddot{\phi}^2 =& r^2\frac{d}{dt}\dot{\theta} \\
-r\sin(\theta)(R + r\cos(\theta))\ddot{\phi}^2 =& r^2\ddot{\theta}
\end{aligned}$$

1.9.3 Sphere

Let a sphere be parametrized by:

$$\begin{aligned}
r &= (x, y, z) \\
x &= \cos(\theta)\cos(\phi) \\
y &= \cos(\theta)\sin(\phi) \\
z &= \sin(\theta)
\end{aligned}$$

The partials of r are given by:

$$\begin{aligned}
\frac{\partial r}{\partial \phi} &= (-\cos(\theta)\sin(\phi), \cos(\theta)\cos(\phi), 0) \\
\frac{\partial r}{\partial \theta} &= (-\sin(\theta)\cos(\phi), -\sin(\theta)\sin(\phi), \cos(\theta)) \\
\frac{\partial r}{\partial \phi} \cdot \frac{\partial r}{\partial \phi} &= \cos(\theta)^2 \\
\frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \theta} &= 1 \\
\frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \phi} &= 0
\end{aligned}$$

Use the partial to get the line element:

$$\begin{aligned}
\dot{r}^2 &= \left(\frac{\partial r}{\partial \phi}\dot{\phi} + \frac{\partial r}{\partial \theta}\dot{\theta}\right)^2 \\
&= \frac{\partial r}{\partial \phi} \cdot \frac{\partial r}{\partial \phi}\dot{\phi}^2 + \frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \theta}\dot{\theta}^2 + 2\frac{\partial r}{\partial \theta} \cdot \frac{\partial r}{\partial \phi}\dot{\phi}\dot{\theta} \\
&= \cos(\theta)^2\dot{\phi}^2 + \dot{\theta}^2
\end{aligned}$$

Geodesic Equations:

$$\begin{aligned}
\ddot{\theta} &= -\sin(\theta)\cos(\theta)\dot{\phi}^2 \\
0 &= \frac{d}{dt}\cos(\theta)^2\dot{\phi} \\
&= \cos(\theta)^2\ddot{\phi} - 2\cos(\theta)\sin(\theta)\dot{\phi}\dot{\theta}
\end{aligned}$$

Consider the function:

$$\begin{aligned}
f(t) &= z + \alpha x + \beta y \\
f(t) &= \sin(\theta) + \alpha\cos(\theta)\cos(\phi) + \beta\cos(\theta)\sin(\phi) \\
\dot{f}(t) &= \dot{\theta}\cos(\theta) + \alpha(-\dot{\theta}\sin(\theta)\cos(\phi) - \dot{\phi}\cos(\theta)\sin(\phi)) + \beta(-\dot{\theta}\sin(\theta)\sin(\phi) + \dot{\phi}\cos(\theta)\cos(\phi))
\end{aligned}$$

1.10 Cosine Transform

Let $A(\xi)$ and $\phi(\xi)$ be given amplitudes and phase offset for a cosine transform:

$$\int_{-\infty}^{\infty} A(\xi)\cos(2\pi(t\xi+\phi(\xi)))\,d\xi$$

I like that the functions are fully real, but we can convert this form to a regular Fourier transform through:

$$\begin{aligned}
&\int_{-\infty}^{\infty} A(\xi)\cos(2\pi(t\xi+\phi(\xi)))\,d\xi \\
=&\int_{-\infty}^{\infty} A(\xi)\frac{1}{2}\big[\exp(2\pi i(t\xi+\phi(\xi)))+\exp(-2\pi i(t\xi+\phi(\xi)))\big]\,d\xi \\
=&\int_{-\infty}^{\infty} \frac{A(\xi)\exp(2\pi\phi(\xi))-A(-\xi)\exp(2\pi\phi(-\xi))}{2}\exp(2\pi it\xi)\,d\xi
\end{aligned}$$

Addition of exponentials:

$$\begin{aligned}
\exp(a)+\exp(b) =& \Big[\exp(xa)+\exp(xb)\Big]_{x=1} \\
=&2\pi\left[\int_{-\infty}^{\infty}\big(\delta(k-a)+\delta(k-b)\big)\exp(-ixk)\,dk\right]_{x=1} \\
&\text{By letting } k=k'+\frac{a+b}{2} \\
=&\exp\left(\frac{a+b}{e}\right)\left(\exp\left(\frac{a-b}{e}\right)+\exp\left(\frac{b-a}{e}\right)\right)
\end{aligned}$$

Observe how the second line's form resembles the conversion between Fourier and cosine transforms.

1.11 Cool Chebyshev Generating Functions

1.11.1 Elementary Generating Functions

$$\begin{aligned}\sum_{i=0}^{\infty} T_n(x)t^n =& \frac{1-xt}{1-2xt+t^2}\\ \sum_{i=0}^{\infty} U_n(x)t^n =& \frac{1}{1-2xt+t^2}\end{aligned}$$

With

$$\begin{aligned}T_n(\cos(\theta)) =& \cos(n\theta)\\ U_n(\cos(\theta))\sin(\theta) =& \sin((n+1)\theta)\end{aligned}$$

For example:

$$\begin{aligned}\sum_{n=0}^{\infty} t^n \sum_{k=0}^{n} T_k(\cos(\theta)) =& \frac{1}{1-t}\cdot\frac{1-\cos(\theta)t}{1-2\cos(\theta)t+t^2}\\ =& \frac{1-t\cos(\theta)}{\sin(\theta)(1-t)}\cdot\frac{\sin(\theta)}{1-2\cos(\theta)t+t^2}\\ =& \frac{1-t\cos(\theta)}{\sin(\theta)(1-t)}\sum_{n=0}^{\infty} t^n \sin((n+1)\theta)\end{aligned}$$

1.11.2 Complex Exponential

There's some stuff you can do with the complex exponential.
First is an explanation for the generating function forms:

$$\begin{aligned}\sum_{n=0}^{\infty} t^n \exp(in\phi) =& \frac{1}{1-t\exp(i\phi)}\\ =& \frac{1-t\exp(-i\phi)}{1+t^2-2t\cos(\phi)}\\ =& \frac{1-t(\cos(\phi)-i\sin(\phi))}{1+t^2-2t\cos(\phi)}\\ =& \sum_{n=0}^{\infty} t^n\cos(n\phi) + i\sum_{n=0}^{\infty} t^{n+1}\sin((n+1)\phi)\end{aligned}$$

But also stuff like:

$$\begin{aligned}\sum_{n=0}^{\infty} nt^n \exp(in\phi) =& \frac{t\exp(i\phi)}{(1-t\exp(i\phi))^2} \\ =& \frac{t\exp(i\phi)(1-t\exp(-i\phi))^2}{(1+t^2-2t\cos(\phi))^2} \\ =& \frac{t\cos(\phi)(1-2t\cos(\phi)+t^2)+t\sin(\phi)(1-2t\sin(\phi)-t^2)}{(1+t^2-2t\cos(\phi))^2}\end{aligned}$$

The notes kind of end here and I don't remember what I was cooking. The denominator and the $t\cos(\phi)$ factor resemble the generating function but need more massaging.

1.11.3 Orthogonality

Let:

$$f(x,t) = \frac{1-t^2}{1-2tx+t^2}$$

We have:

$$\int_{-1}^{1} T_n(x) f(x,t) \frac{dx}{\sqrt{1-x^2}} \quad = \pi t^n$$

Let P be some polynomial:

$$P(x) = \sum_n p_n T_n(x)$$

Then:

$$P(x) = (1-2tx+t^2)Q(x,t) + P\left(\frac{1+t^2}{2t}\right)$$

For some polynomial Q with coefficients dependent on t.
Let $q(t)$ be the coefficient from polynomial $T_0(x)$.
Then:

$$\pi \sum_n p_n t^n = \int_{-1}^{1} P(x) f(x,t) \frac{dx}{\sqrt{1-x^2}} = \pi q(t)(1-t^2) + P\left(\frac{1+t^2}{2t}\right)$$

Derive and sub $t = \pm 1$ to get:

$$\sum_n n p_n = -2q(1)$$

$$\sum_n np_n(-1)^n = 2q(-1)$$

Might work for some general function, not sure.

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2.1 Interest Identities

Let P be the principle invested at a rate of r. Consider four different investment scenarios:

- Not invested: $P_0 = P$.
- Fully Invested at the beginning, one instalment at the end:

$$P_1 = (1 + r)P$$

- Continuously invested, continuous installments:

$$P_2 = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{P}{n} \left(1 + \frac{r}{n}\right)^k = \frac{\exp(r) - 1}{r} P$$

- Fully Invested at the beginning, continuous instalments:

$$P_3 = \lim_{n\to\infty} P \left(1 + \frac{r}{n}\right)^n = \exp(x)P$$

Interestingly the relative size of P_2 and P_1 depend on r. P_1 starts on P_2 but switches as r increases.

P_3 is always the best, the proof for P_1 and P_0 are obvious. $P_3 > P_2$ follows from:

$$0 < \int_0^r t \exp(t)\, dt = \big[(t-1)\exp(t)\big]_0^r = (r-1)\exp(r) + 1$$

The following interesting identities hold:

$$P_3 = rP_2 + P_0$$

$$P_3 - P = r(P_2 - P) + (P_1 - P)$$

The breaks first neatly breaks P_3 into a nice linear sum. The second does similar for the profit of the investment, total yield minus principle.

2.2 Lagrange Multiplier

Recall that local extrema x of the function $f : \mathbb{R}^n \to \mathbb{R}$ subject to contrasts g_i satisfy:

$$\nabla f(x) = \sum_i \lambda_i \nabla g_i(x)$$

The core observation is that if ∇f has a component outside the span of ∇g_i then you can move in that direction while keeping g_i's constant, contradicting the point being an extrema.

But the actual constants λ_i have a useful interpretation as the rate the value of f at the extrema changes as the constant g_i changes. To see this pick a particular g_j and construct a d such that:

$$d \cdot \nabla g_i = D\delta_{i,j}$$

You can achieve this by iteratively removing components in some matter like the following:

$$d_0 = \nabla g_0,\ d_{n+1} = d_n - d_n \cdot \nabla g_n$$

Now scale d down such that functions around the extrema can be approximated through targets[1]. We have:

$$\begin{aligned}
g_i(x+d) =& g_i(x) + d \cdot \nabla g_i(x) \\
=& g_i(x) + D\delta_{i,j} \\
f(x+d) =& f(x) + d \cdot \nabla f(x) \\
=& f(x) + \sum_i \lambda_i d \cdot \nabla g_i(x) \\
=& f(x) + D\lambda_j \\
\nabla f(x+d) =& \nabla(f(x) + D\lambda) \\
=& \nabla f(x) \\
=& \sum_i \lambda_i \nabla g_i(x) \\
=& \sum_i \lambda_i \nabla\big(g_i(x+d) - D\delta_{i,j}\big) \\
=& \sum_i \lambda_i \nabla g_i(x+d)
\end{aligned}$$

[1]Those so inclined are free to chase $\epsilon - \delta$'s

From the these equation we can see that $x + d$ satisfy the requirement to be an extrema. We can also see that a change of D in g_i created a change of $\lambda_j D$ in the value at the extrema, hence giving a rate of change of λ_j.

To-Do: Add and example of minimizing height when the two contrasts are parabolic and linear. (You will need to use logs to get the change for the parabola to be a change in it's width and not height.)

2.3 Isosets

Consider the function $f : \mathbb{R}^n \to \mathbb{R}$ let the isosets[2] be sets S in the domain of f such that $f(S)$ is constant.

Given some isoset S and point $x \in \mathbb{R}^n$ of f we want some kind of function that returns some type of measure $d \in \mathbb{R}$ of the distance of x to S. This function will be used in fragment shader rendering, which is why the mission statement is so vague. We only need some general measure since it's better to efficiently get that measure and tweak coefficients then to get something perfectly accurate.

A particular application is the $n = 2$ case where we are looking to find the distance for the contour line.

2.3.1 Naïve Solution

The first idea is to compare the distance of $f(x)$ to $f(S)$ to d:

$$d = |f(x) - f(S)|$$

The problem with this solution is that the measure changes as a function of $|\nabla f(x)|$. That is to say that the faster f changes at x the closer x has to be to S to get the same value of d.

This method might work for some shader effects but not others. For example it won't work to draw a constant width contour as the width would be inversely proportional to $|\nabla f(x)|$

2.3.2 Better Solution

If the underestimation is proportional to $|\nabla f(x)|$ the obvious solution is to divide by $|\nabla f(x)|$:

$$d = \frac{|f(x) - f(S)|}{|\nabla f(x)|}$$

This is the solution currently used but deserves more analysis.

[2]Not the proper name, but I can't recall the proper name right now.

For starters consider the case that x is near the point $s \in S$. Then $\nabla f(s)$ points away from S, that is that if a tangent to S exists at s then $\nabla f(s)$ is at orthogonal to the tangent, by definition of S being a set such that f is constant. The combination of orthogonality and closeness lets us recover the original expression by use of the tangent surface:

$$\begin{aligned}
f(x) =& f(s) + (x - s) \cdot \nabla f(s) \\
|f(x) - f(s)| =& |(x - s) \cdot \nabla f(s)| \\
=& |x - s||\nabla f(s)| \\
\frac{|f(x) - f(s)|}{|\nabla f(s)|} =& |x - s|
\end{aligned}$$

Now consider the case where x is not close to S and the use case of drawing a constant width contour line. Under what conditions do we avoid a false positive? (That is the function thinks x is closer than it is.) Well we need some constraints on the rate at which $\nabla f(x)$ can grow. To see this consider a point far away from S but whose rate of change is very slow between most of x and s, so that $|f(x) - f(s)|$ is small, but suddenly increases at x, such that $|\nabla f(x)|$ is large. This causes their ratio to be small despite $|x - s|$ being large, false saying x should be colored as part of the contour line. If we reverse the set up, rate of change is large at first then slow, we will get a false negative. (That the point is further than we think it is)

To see how a constraint would be useful, consider the following one:

$$|\nabla f(x + d)| \leq k|d||\nabla f(x)|$$

$$
\begin{aligned}
|f(x) - f(s)| =& \left| \int_0^1 \nabla f(x + (s-x)t) \cdot (s-x)\, dt \right| \\
=& \left| (s-x) \cdot \int_0^1 \nabla f(x + (s-x)t)\, dt \right| \\
\leq& |(s-x)| \left| \int_0^1 \nabla f(x + (s-x)t)\, dt \right| \\
& \text{Cauchy-Schwartz} \\
\leq& |(s-x)| \int_0^1 |\nabla f(x + (s-x)t)|\, dt \\
& \text{ML Bound} \\
\leq& |(s-x)| \int_0^1 k|(s-x)t||\nabla f(x)|\, dt \\
\frac{|f(x) - f(s)|}{|\nabla f(x)|} \leq& \frac{k}{2}|s-x|^2
\end{aligned}
$$

This avoids a false negative as x must be at least $\sqrt{\frac{2d}{k}}$ away.

To-do: we need a inequities like $d \geq p(|x-s|)$ to get a bound on false positives.

The condition:

$$|\nabla f(x+d)| \leq k|d| + |\nabla f(x)|$$

Gives:

$$d \leq |x-s| \left(1 + \frac{|x-s|}{|\nabla f(x)|}\right)$$

2.4 Padé Approximant

We wish to approximate a function f by creating a rational function that agrees with f's first N derivatives at zero.

Let T_N be the Nth degree Maclaurin series of f. Consider the steps of finding the polynomial greatest common division by the extended Euclid algorithm of T_N with x^{N+1} where we prematurely stop:

$$\begin{aligned}
x^{N+1} =& 1 \cdot x^{N+1} + & 0 \cdot T_N(x) \\
T_N(x) =& 0 \cdot x^{N+1} + & 1 \cdot T_N(x) \\
r_1(x) =& 1 \cdot x^{N+1} + & -q_1(x) \cdot T_N(x) \\
& \vdots & \\
P(x) =& K(x) x^{N+1} + & Q(x) T_N(x)
\end{aligned}$$

By inspection we get the useful relation:

$$P(x)/Q(x) \equiv T_N(x) \mod x^{N+1}$$

Hence satisfying the original objective. Note that we can chose decrease the degree of P by simply continuing the algorithm.

2.4.1 The Reverse

Say I want to do the reverse, that I have $f = g/h$ where I have the power series for g and h but want to effectively find f We have:

$$\begin{aligned}
x^{N+1} =& 0 \cdot f(x) + & 1 \cdot x^{N+1} \\
g(x) =& h(x) \cdot f(x) + & 0 \cdot x^{N+1}
\end{aligned}$$

You can remove some higher term of f into a residue function K on x^{N+1}, since we don't really care about it.

2.4.2 Differential

What if g and h are related buy a differential equation? We can use it like how we used the quotient-remainder equation. The derivative is linear after all.

2.4.3 Chinese Remainder Theorem

Since I have modulo relations can I combine them with the Chinese Remainder Theorem? The moduli will have to be pairwise coprime $(x - a_i)^n$ stand out.

2.5 p-adic numbers

2.5.1 Valuation

A function v of a field is called a valuation if:

$$\begin{aligned} v(x) =& \infty \quad \text{iff } x = 0 \\ v(xy) =& v(x) + v(y) \\ v(x+y) \geq& \min(v(x), v(y)) \quad \text{with equality if } v(x) \neq v(y) \end{aligned}$$

(Note that the codomain is only required to be an abelian totally ordered group extended with ∞, But I will treat is as the natural numbers.)

There are three immediate corollaries of the definitions.
1:By induction on the inequality we have:

$$v\left(\sum_k x_k\right) \geq \min\left(\bigcup_k \{v(x_k)\}\right)$$

2: By setting $x = y = 1$ in the second equality we have $v(1) = 2v(1)$ and hence $v(1) = 0$.
3: By setting $xy = 1$ in the second equality we have $v(1/x) = -v(x)$.

2.5.2 p-adic numbers

Interestingly, the amount of times a prime p divides a rational number x is a valuation. Let $x = p^n \frac{a}{b}$ where a and b are coprime, then $v_p(x) = n$ is a valuation. (Assuming you set $v_p(0) = \infty$). This is easy to prove, if you need help remember that for a prime p we have: $p|ab \Rightarrow p|a$ or $p|b$.

The reason this is interesting is because $|x|_p = p^{-v_p(x)}$ is a metric on $\mathbb{Q}$. Meaning we can make a new field $\mathbb{Q}_p$ by taking all the limits in $\mathbb{Q}$ as we would nomarlly do to make $\mathbb{R}$ with $|\cdot|$. And through something called Ostrowski's theorem becomes a lot more motivated and less arbitrary way to complete $\mathbb{Q}$.

But the valuation alone also provides two cool proofs that simplify previous proofs.

2.5.3 Irrationality of $\sqrt{2}$

Consider the equation:

$$x^2 = 2$$

Valuating both sides gives:

$$2v_2(x) = 1$$

But $v_2(x)$ is an integer for all rational x hence the LHS is always even but the right is odd. Hence there is no x satisfying the equation and $\sqrt{2}$ is irrational.

2.5.4 Valuation of the Harmonic Numbers

The valuation of harmonic numbers is given by $v_2(H_n) = -\lfloor \log_2(n) \rfloor$.

Lemma: Let k be the power of the largest power of 2 less than n, i.e. $k = \lfloor \log_2(n) \rfloor$. Let S be the set $[1, n]$ excluding 2^k, then from the maximality of k we have:

$$\max(v_2(S)) \leq k - 1$$

Since if we assume there is an $s \in S$ such that $v_2(s) > k-1$ with the codomain of v_2 being integers means $v_2(s) \geq k$. This means there exists an integers a and b coprime to each other and 2 such that $s = 2^k \frac{a}{b}$. s being a positive integer means $b = 1$. $a \neq 1$ since it would make $s = 2^k$, which was excluded from S. But $a \geq 2$ would means $2^{k+1} \in S$, contradicting the maximality of k.

Now H_n can be written as the following sum:

$$H_n = \sum_{s \in S} \frac{1}{s} + 2^{-k}$$

Where the valuation of first term is bound by:

$$\begin{aligned} v_2\left(\sum_{s \in S} \frac{1}{s}\right) &\geq \min\{v_2(1/s) \mid s \in S\} \\ &= \min\{-v_2(s) \mid s \in S\} \\ &= -\max\{v_2(s) \mid s \in S\} \\ &= -k + 1 \end{aligned}$$

Hence the valuations are not equal since:

$$v_2(2^{-k}) = -k < -k + 1 \leq v_2\left(\sum_{s \in S} \frac{1}{s}\right)$$

Hence

$$v_2(H_n) = \min\left\{v_2\left(\sum_{s \in S} \frac{1}{s}\right), v_2(2^{-k})\right\} = -k$$

As required.

Note that for $n \geq 2$ we have $k \geq 1$ and hence $v_2(H_n) \leq -1$ meaning H_n isn't an integer for $n \neq 1$

2.5.5 Valuation of the Harmonic Numbers v2

I think the lemma's of the previous proof can be separated and cleaned up.

Lemma: Let S be a subset of the v's domain with an element $s_0 \in S$ such that:

$$v(S \backslash \{s_0\}) > v(s_0)$$

Then:

$$v\left(\sum_{s \in S} s\right) = v(s_0)$$

Proof: Plugging the inequality into the valuation inequality axiom gives:

$$v\left(\sum_{s \in S \backslash \{s_0\}} s\right) \geq v(S \backslash \{s_0\}) > v(s_0)$$

Hence the term of the far left and far right are not equal meaning:

$$\begin{aligned} v\left(\sum_{s \in S} s\right) =& v\left(s_0 + \sum_{s \in S \backslash \{s_0\}} s\right) \\ =& \min\left\{v(s_0), v\left(\sum_{s \in S \backslash \{s_0\}} s\right)\right\} \\ =& v(s_0) \quad \square \end{aligned}$$

Lemma: If $n \in \mathbb{N}$ then $v_p(n) \leq \log_p(n)$ with equality iff n is a power of p.

Proof: Equality in the case of n being a power is trivial. Now consider n not a power of 2 meaning $n = a\, p^{v_p(n)}$ where $a > 1$, hence:

$$p^{v_p(n)} < n = p^{\log_p(n)}$$

$\log_p$ is strictly monotonic, hence:

$$v_p(n) < \log_p(n)$$

Lemma: If $n \in \mathbb{N}$ then $v_2(n) \leq \lfloor \log_2(n) \rfloor$ with equality iff n is a power of 2.

Proof: Equality in the case of n being a power is trivial. Assume n isn't a power of 2 then:

$$n \geq 3 \cdot 2^{v_2(n)}$$

Hence:

$$\begin{aligned} \log_2(n) \geq& \log_2(3) + v_2(n) \\ \log_2(n) - \log_2(3) \geq& v_2(n) \\ \lfloor \log_2(n) - \log_2(3) \rfloor \geq& \lfloor v_2(n) \rfloor \end{aligned}$$

Since $\log_2(3) > 1$ we have:

$$\lfloor \log_2(n) \rfloor > \lfloor \log_2(n) - \log_2(3) \rfloor$$

And $v_2(n)$ is an integer, hence: $\lfloor \log_2(n) \rfloor > v_2(n) \quad \square$

Note that this can't be generalized to larger p since the logarithm being less than one isn't guaranteed. Compare with $p = 3$ with $v_3(6) = 1 = \lfloor \log_3(6) \rfloor$ because $\log_3(2) < 1$. Note to the previous note, in this case you can say $v_3(n) \leq \lfloor \log_p(n) \rfloor$ iff n is a power of 3 or even, which is still *kind of* cool.

Lemma: Let $S_n = \{k^{-1} | 1 \leq k \leq n\}$ and $k_0 = \lfloor \log_2 n \rfloor$. Then $2^{-k_0} \in S$ and satisfies:

$$v_2(S_n \backslash \{2^{-k_0}\}) > v_2(2^{-k_0})$$

Proof: $2^{k_0} = 2^{\lfloor \log_2 n \rfloor} \leq 2^{\log_2 n} = n$ hence $2^{-k_0} \in S$. Now consider $s \in S$ then $s^{-1} \in \mathbb{N}$ which from the previous lemma means:

$$v_2(s^{-1}) \leq \lfloor \log_2(s^{-1}) \rfloor \leq \lfloor \log_2(n) \rfloor = k_0$$

With equality iff s^{-1} is a power of 2. It's can't be a larger power than 2^{k_0} since $2^{\lfloor \log_2(n) \rfloor}$ is the largest power less than or equal to n. But from the definition of $S_n \backslash \{2^{-k_0}\}$ it can't be the same power. Hence s^{-1} can only be a lower power making the final inequality strict anyway. Hence:

$$v_2(s^{-1}) < k_0 \Rightarrow v_2(s) > -k_0 = v_2(2^{-k_0}) \quad \square$$

Noting that $H_n = \sum_{s \in S_n} s$ means the proof that $v_2(H_n) = -\lfloor \log_2(n) \rfloor$ follows immediately from the first and last lemmas.

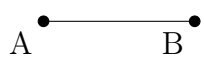
2.6 Quick Summary of Spaces

A space is a collection of elements, often called points, with some additional structure. There are four main types of spaces that form a nice hierarchy, that is that some types of spaces are always a subtype of other space.

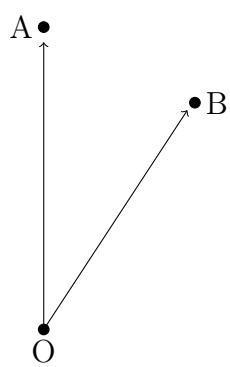
2.6.1 Hierarchy

Topological Space: Neighbourhood In a topological space the elements have a concept of "Neighbourhood", that is the points which are adjacent to each point.

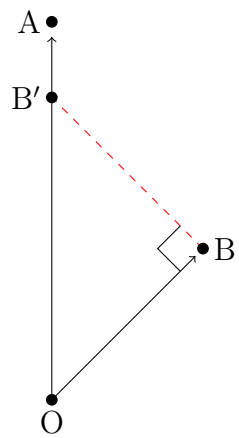
Metric Space: Distance In a metric space the element have a concept of distance to each other. You can induce a neighbourhood by saying all points within a certain threshold distance to a point are in the neighbourhood of that point.



Normed Space: Length In a normed space there is a concept of length of a point. Shown here as their distance to some origin. A distance can be induced by getting the length of an arrow going from A to B.



Inner-product Space: Projection An inner product space has a concept of projecting one element on another. You can induce a length by projecting an object onto itself.



2.6.2 Euclidean

Euclidean space is a normed space on $\mathbb{R}^n$ where the inner product is given by $\sum_k A_k B_k$. This naturally induces a metric on $\mathbb{R}^n$.

The reason we care about the metric of Euclidean space is because all normed spaces have a form of the Pythagorean theorem. Making metric space the most endowed space we can easily start thinking about non-Euclidean space.

This sucks because most talk about non-Euclidean space talks about "parallel lines" which makes one naturally think about angles and hence an inner product. But no, instead we have the concept of a geodesic. A geodesic is a curve that is locally distance minimizing. This locality comes from the topology, and means that for all points $\gamma(t_1)$ and $\gamma(t_0)$ on the curve γ such that they are in the same neighbourhood satisfy:

$$d(\gamma(t_1), \gamma(t_0)) = k|t_1 - t_2|$$

2.6.3 non-Euclidean

Quick rundown of some attempts of non-Euclidean geometries:

Pseudo-Euclidean: Like how we defined Euclidean Space was defined with an inner product we define a new structure with a different quadratic form.

Riemann Manifold: A Manifold is a space that is "locally Euclidean". And a Riemann Manifold is one where this inner-product in the local Eu-

clidean space is always positive.

Pseudo-Riemann Manifold: Like A Riemann Manifold but the inner-product is only required to be non-degenerate. I think this includes Pseudo-Euclidean space, but haven't put much thought into it.

2.7 XOR hash

Today I was presented with the following problem:

Given an array with all integers $[1, 100]$ with a single integer removed. Assuming memory is limited, how do you efficiently determine the missing integer?

The solution is to XOR all the elements of the array together. Then XOR this with the known value for ALL the numbers between $[1, 100]$. The result will be the missing number.

This problem seemed like a good introduction to hashes, like Zobrist, that XOR a bunch of info together. The benefit of this approach is that the components can be added/removed by a XOR:

$$h(\{a_0, a_1\}) = h(\{a_0\}) \operatorname{XOR} h(\{a_1\})$$

2.7.1 Determining the constant

It turns out its easy to calculate the constant for the original question by hand. First define T as:

$$T(n) = n \operatorname{XOR} T(n-1), \quad T(1) = 1$$

Then:

$$T(n) = \begin{cases} n & n \mod 4 = 0 \\ 1 & n \mod 4 = 1 \\ n+1 & n \mod 4 = 2 \\ 0 & n \mod 4 = 3 \end{cases}$$

This can easily be proved by induction and remembering that for even n we have:

$$n \operatorname{XOR} 1 = n + 1$$

Because of the modularity you would expect there to be an expression for T involving powers of the fourth root of unity (i):

$$T(n) = \frac{1}{2} + \frac{1}{2}(1 + (-1)^n) + \frac{1}{4}((i-1)(-i)^n - (i+1)i^n)$$

(Done on scrap paper, not double checked, close enough to see the form).

While the hybrid function is probably a more useful form the last one involves complex numbers in a way I didn't expect.

2.7.2 Homomorphic Hashing

Those with mathematic training will notice that the property that make this approach work:

$$h(\{a_0, a_1\}) = h(\{a_0\}) \operatorname{XOR} h(\{a_1\})$$

Has the same form as a homomorphism[3] and would investigate homomorphic hash functions.

Well the most influential definition of such a function happened in 2004 Krohn, Freedman and Mazieres proposed a definition of homomorphic hash function as function $H : V \to G$ such that:

- H is collision resistant. Meaning we are unlikely to find x and y such that $H(x) = H(y)$.
- H is a homomorphism. Meaning $H(x+y) = H(x)+H(y)$ for all x and y.

The main problem is defining V that is compatible with the regiments on H. Since we want to use the set union operator then the natural choice is $V = 2^S$ for some set S. But our hash only has the same form as a homomorphism[4] when the input sets are mutually exclusive. In general we have:

$$H(S_0 \cup S_1) = H(S_0) \operatorname{XOR} H(S_1) \operatorname{XOR} H(S_0 \cap S_1)$$

We can make a collision resistant hash though. Make H uniformly distributed on $\mathbb{F}_2^{\log_2 |S|}$ for S. Then by induction and $x \operatorname{XOR} y = 0$ iff $x = y$ you can show that it stays uniform for 2^S. Meaning the chance of a collision is $\frac{1}{|S|}$

(This is actually a bound for when $|S|$ is a power of 2, but you can figure out the rest)

Homomorphic functions are still useful in cryptography though.

[3] "same form" as a "homomorphism", math pun intended

[4] haha

2.8 Jensen's Inequality

A convex function ϕ on a set X is one such that for $a_n \in X$ with $w_0 + w_1 = 1$ and $w_n \geq 0$ then we have:

$$\phi(w_0 a_0 + w_1 a_1) \leq w_0 \phi(a_0) + w_1 \phi(a_1)$$

Jensen's Inequality states that expected value of the function is less then the function of the expected value:

$$\phi(\mathbb{E}(X)) \leq \mathbb{E}(\phi(X))$$

By multiplying the function by -1, we get the intuitive corollary that the inequality is reversed for concave functions.

2.8.1 Generalization

A more general form of convexity can be easily proved by induction.

Assume that for some n we have that for all $\sum_{i=1}^{n} w_i' = 1$ and $w_i' \geq 0$:

$$\phi\left(\sum_{i=1}^{n} w_i' a_i\right) \leq \sum_{i=1}^{n} w_i' \phi(a_i)$$

Let $\sum_{i=1}^{n+1} w_i = 1$ and $W = \sum_{i=1}^{n} w_i$ we have $w_{n+1} + W = 1$ and :

$$\begin{aligned}
\phi\left(\sum_{i=1}^{n+1} w_i a_i\right) =& \phi\left(w_{n+1} a_{n+1} + W \sum_{i=1}^{n} \frac{w_i}{W} a_i\right) \\
\leq& w_{n+1} \phi(a_{n+1}) + W \phi\left(\sum_{i=1}^{n} \frac{w_i}{W} a_i\right) \\
\leq& w_{n+1} \phi(a_{n+1}) + W \sum_{i=1}^{n} \frac{w_i}{W} \phi(a_i) \\
\leq& \sum_{i=1}^{n+1} w_i \phi(a_i)
\end{aligned}$$

Viewing the weights as probabilities this form can directly be in interpreted as a discrete form of Jensen's Inequality.

2.8.2 Mean Inequalities

This can be directly applied to various mean inequalities.

$$AM = \frac{1}{n}\sum_{i=1}^{n} a_i$$

$$RMS = \sqrt{\frac{1}{n}\sum_{i=1}^{n} a_i^2}$$

$$GM = \sqrt[n]{\prod_{i=1}^{n} a_i}$$

Using $\phi(x) = x^2$,$w_i = \frac{1}{n}$ we directly get:

$$AM^2 \leq RMS^2$$

Using $\phi(x) = \log(x)$,$w_i = \frac{1}{n}$ we get: [5]

$$\log(AM) \geq \log(GM)$$

Since both these functions are also monotonic the clean relations follow.

2.8.3 Decision Theory

Interpret ϕ as a utility function then the convexity tells us whether we take the fixed $\mathbb{E}(X)$ or risk X. Well if ϕ is convex then:

$$\phi(\mathbb{E}(X)) \leq \mathbb{E}(\phi(X))$$

And we should take the risk, there is more expected utility taking the risk then the utility of $\mathbb{E}(X)$.

If ϕ is concave then we should not take the risk.

[5] Remember that the inequality is reversed since log is concave

2.9 Convex

TODO: Flesh out, copy edit, and maybe include pictures for the geometric intuition in the cord section.

In the section of Jensen's Inequality I defined a convex function ϕ on a set X as one such that for $a_n \in X$ with $w_0 + w_1 = 1$ and $w_n \geq 0$ then we have:

$$\phi(w_0 a_0 + w_1 a_1) \leq w_0 \phi(a_0) + w_1 \phi(a_1)$$

I've thought of another proof of the previous section but iterates over functions instead of points. And we can use some geometry intuition (cords).

2.9.1 Cords

First let me define a new function called the cord function that takes on an interval $[a, b] \subseteq X$:

$$C_{[a,b]}(x) = \phi(a) + (x - a)\frac{\phi(b) - \phi(a)}{b - a}$$

The convex property can be changed to:

$$x \in [a, b] \Rightarrow \phi(x) \leq C_{[a,b]}(x)$$

Lemma:, if $x_0 \leq x_1 \leq x_2$ then:

$$x \in [x_0, x_1] \Rightarrow C_{[x_0,x_1]}(x) \leq C_{[x_0,x_2]}(x)$$

and:

$$x \in [x_1, x_2] \Rightarrow C_{[x_1,x_2]}(x) \leq C_{[x_0,x_2]}(x)$$

Proof: Use $C_{[a,b]}(a) = \phi(a)$ or $C_{[a,b]}(b) = \phi(b)$ and $x_1 \in [x_0, x_2]$ with the definition of convexity.

Lemma:, if $a_0 \leq a_1 \leq b_1 \leq b_0$ then:

$$x \in [a_1, b_1] \Rightarrow C_{[a_1,b_1]}(x) \leq C_{[a_0,b_0]}(x)$$

Proof: use the previous lemma with the triples form the quad inequality, then chain together the newer inequalities.

Let:

$$\phi_{[A,B]}(x) = \begin{cases} C_{[A,B]}(x) & x \in [A, B] \\ \phi(x) & \text{else} \end{cases}$$

$\phi_{[A,B]}(x)$ is convex. Prove this using the previous lemmas with the three cases (points same side, both sides, one in one out).

2.9.2 The Cool Proof

We want to show that for all $\sum_{i=1}^{n} w_i' = 1$ and $w_i' \geq 0$:

$$\phi\left(\sum_{i=1}^{n} w_i' a_i\right) \leq \sum_{i=1}^{n} w_i' \phi(a_i)$$

Assume it works for $n-1$ points. Take two points (w_0, a_0) and (w_1, a_1) and make a new convex function using the method from the previous section with $[x_0, x_1]$, then remove the two points and replace with $(w_0 + w_1, \frac{w_0 a_1 + w_1 a_1}{w_1 + w_0})$. Now we have a $n-1$ points, and if you expand the algebra you get the correct form, hence by induction you prove the initial result.

2.10 Isomorphisms of the quotient rings of $\mathbb{Z}[X]$

While working elsewhere I thought that it's actually really easy to prove:

$$\gcd(a, d) = \gcd(b, d) \Rightarrow \exists z \in \mathbb{Z} \text{ such that } z\frac{a}{d} - \frac{b}{d} \in \mathbb{Z}$$

Using Bézout's identity.

2.10.1 The proof

From Bézout's identity we know there exits r, s in $\mathbb{Z}$ such that:

$$ra + sd = \gcd(a, d) = \gcd(b, d)$$

From the definition of gcd there exists $d' \in \mathbb{Z}$ such that:

$$b' \gcd(b, d) = b$$

Multiplying the first equation by b' gives:

$$(rb')a + (sb')d = b' \gcd(b, d) = b$$

Hence:

$$(rb')\frac{a}{d} - b\frac{b}{d} = -(sb')$$

As required.

Note that we can actually control the size, and hence sign, of z by using the:

$$ra + sd = a(r + bk) + b(s - ak)$$

Trick, meaning we can force $z > 0$ and hence $z \in \mathbb{N}$.

2.10.2 The questions

So this proof brings up some questions:

1. Doesn't this prove that $\mathbb{Z}[X]/(dX - a) \cong \mathbb{Z}[X]/(dX - b)$?
2. Couldn't this be generalized to Bézout Domains?
3. What about the converse?

2.10.3 Question 1

Let:

$$A = \mathbb{Z}[X]/(dX - a), \quad B = \mathbb{Z}[X]/(dX - b)$$

We can say A and $B \subset \mathbb{Q}$ where we identity X with $\frac{a}{d}$ and $\frac{b}{d}$ respectfully. But since we can find a linear relation between the two Xs in terms, we need two relations but we can avoid division completely, of the sub-field $\mathbb{Z}$ I'd expect an isomorphism.

Assume the gcd condition is meet and let z_0 and z_1 be the integers such that:

$$z_1 \frac{a}{d} - \frac{b}{d} = -z_0$$

Equally written as:

$$d(z_1 a + z_0) = b$$

Attempt 1

Define $\phi : B \to A$. A ring homomorphism fixes 1 and hence acts identically on the set generated by 1, that is $\mathbb{Z}$. Since $a \in \mathbb{Z}$ hence:

$$\begin{aligned} a =& \phi(a) \\ =& \phi(dX) \\ =& d\phi(X) \end{aligned}$$

Suggesting a form for $\phi(X)$:

$$\phi(X) = z_1 X + z_0$$

This is all the degrees of freedom we get since being a homomorphism requires:

$$\phi\left(\sum_n p_n X^n\right) = \sum_n \phi(p_n)\phi(X)^n$$

Hence the suggested form is:

$$\phi\left(\sum_n p_n X^n\right) = \sum_n p_n (z_1 X + z_0)^n$$

But dealing with showing homomorphism when dealing the canceling term isn't something I want to do.

Attempt 2

Define a function $\phi : \mathbb{Z}[X] \to B$ where:

$$\phi\left(\sum_n p_n X^n\right) = \sum_n p_n (z_1 X + z_0)^n \mod (dX - b)$$

Then we can use the first isomorphism theorem of rings and only need to show that ϕ is surjective and has a kernel of $\langle dX - a \rangle$.

2.11 Generated Subfield of Finite Fields

Let the finite field F have a set of elements S. The set $\langle S \rangle$ whose elements are the sums of the products of powers of S:

$$\langle S \rangle = \left\{ \sum_n \prod_i s_i^{p_{i,n}} \mid s_i \in S \text{ and } p_{i,n} \in \mathbb{Z} \right\} \cup \{0\}$$

For example if $S = \{s_0, s_1, s_2\}$ all of the following are elements of $\langle S \rangle$:

$$s_0 + s_1 + s_2, \quad s_0 s_1 s_2 s_3, \quad 1 + s_0 + s_1 s_2^7 + s_0 s_1 s_2^2$$

(I use the $\langle \cdot \rangle$ notation because this is like generation, one of the most irregular and abused words in algebra).

The set inherits it's operations from F and is clearly closed under addition and multiplication. It also contains 1 and its repeated sum, labeled n, by using $p_{i,n} = 0$:

$$n = \overbrace{(s_0 s_1 s_2)^0 + (s_0 s_1 s_2)^0 + \ldots s(s_0 s_1 s_2)^0}^{n \text{ times}}$$

Given $q \in \langle S \rangle$ because there are a finite number of options for nq the pigeonhole principle means there will eventually be $n > m + 1$ such that:

$$\begin{aligned} nq =& mq \\ (n-m)q =& 0 \\ q + (n-m-1)q =& 0 \end{aligned}$$

$(n - m - 1)q$ which is the additive inverse of q. Because $n - m - 1 > 0$ it is an element of $\langle S \rangle$ and since q is also an element by closure so is the inverse.

Likewise for the multiplicative inverse:

$$\begin{aligned} q^n =& q^m \\ q \cdot q^{n-m-1} =& 1 \end{aligned}$$

Hence $\langle S \rangle$ is a, not necessarily proper, subfield of F.

2.11.1 $S = \{1\}$

An important example is when $S = \{1\}$. Let n be the "lowest"[6] where the pigeonhole principle whole applies:

$$\begin{aligned} m + (n - m) =& n + (m - m) \\ =& n \\ =& m \end{aligned}$$

The minimality[7] of n makes $n-m$ non-zero making n zero. Hence $S \cong \mathbb{F}_n$.

(This is sloppy even for quick notes)

2.11.2 Wedderburn's Little Theorem

This is similar to Wedderburn's little theorem, that all finite domains are fields. Wedderburn generalizes the subject to domains instead of field and thus has to use $q - q^n = 0$ and some polynomial analysis (Headache). But reduces the results, it doesn't discuss closure.

Cool result, I suggest looking it up.

2.11.3 Dreams of a more rigor

Let R be a ring and let ϕ be a function from $\mathbb{Z}$ to R such that:

$$\phi(1) = 1_R, \quad \phi(n \pm 1) = \phi(n) \pm \phi(1_R) = \phi(n) \pm 1_R$$

ϕ is well-defined as you can find the value for any $n \in \mathbb{Z}$ by induction. And the function only has one value at each n since you can't change the value by "doubling back" on the induction:

$$\phi((n \pm 1) \mp 1) = \phi(n \pm 1) \mp 1_R = \phi(n) \pm 1_R \mp 1_R = \phi(n)$$

(This can be made more rigorous by only allowing induction one way in the definition).

[6] Yes I know I haven't defined that properly, but just figure it out.
[7] See previous footnote.

Induction on n shows:

$$\begin{aligned}\phi(n)+\phi(m) =&\phi(n\pm 1)\mp 1+\phi(m)\\ =&\phi(n\pm 1)+\phi(m\mp 1)\\ =&\phi(n\pm 1+m\mp 1)\\ =&\phi(n+m)\end{aligned}$$

Which can be used to further prove:

$$\begin{aligned}\phi(n)\phi(m) =&(\phi(n\pm 1)\mp 1)\phi(m)\\ =&\phi(n\pm 1)\phi(m)\mp\phi(m)\\ =&\phi((n\pm 1)m)\mp\phi(m)\\ =&\phi((n\pm 1)m\mp m)\\ =&\phi(nm)\end{aligned}$$

These identities, combined ϕ being defined with $\phi(1)=1_R$, shows that ϕ is a homomorphism.

Which, by the first isomorphisms theorem for rings, shows that the $\operatorname{img}\phi$ is a subring of R and is isomorphic to $\mathbb{Z}/\ker\phi$. This formalizes the concept of the nth sum of 1_R, and lets us manipulate it better.

Corollary: If R is finite then the additive group of R is also finite. Meaning 1_R has an order, which we label N. In this case $\ker\phi = N\mathbb{Z}$ giving:

$$\operatorname{img}\phi = \frac{\mathbb{Z}}{N\mathbb{Z}}$$

It's really easy to use Bézout's identity on these objects to show which ones are fields.

2.11.4 An Observation on Divisibility

Lemma: $n\in\ker\phi \Rightarrow \ker\phi \supseteq n\mathbb{Z}$

A corollary of ϕ being a homomorphism is that:

$$\begin{aligned}n\in\ker\phi \Rightarrow&\phi(nk)=\phi(n)\phi(k)=0\\ \Rightarrow&n\mathbb{Z}\subseteq\ker\phi\end{aligned}$$

Lemma: If there is a nonzero $n \in \ker\phi$ then there is nonzero $m \in \ker\phi$ such that nonzero $k \in \ker\phi \Rightarrow |k| \geq |m|$

There are finitely many elements with a magnitude less than $|n|$. Hence there are finitely number of element with a magnitude less than $|n|$ that are in the $\ker\phi$. Hence there is an element that takes the minimal value of $|\cdot|$.

Lemma: If there exits nonzero $n \in \ker\phi$ such that nonzero $m \in \ker\phi \Rightarrow |m| \geq |n|$ then $\ker\phi \subseteq n\mathbb{Z}$

(Going forward remember that either $|n| = n$ or $|n| = -n$).

Let $m \in \ker\phi$ since $|m| > |n|$ there exits $q, r \in \mathbb{Z}$ with $|r| < |n|$ such that:

$$|m| = q|n| + r \Rightarrow q|n| + r - |m| = 0$$

But:

$$\begin{aligned} 0 &= \phi(q|n| + r - |m|) \\ &= \phi(q)\phi(|n|) + \phi(r) - \phi(|m|) \\ &= \phi(r) \\ &= \phi(|r|) \end{aligned}$$

Meaning we have $r \in \ker\phi$ and $|r| < |n|$. This requires $r = 0$ otherwise would contradicts the initial assumption, hence:

$$|m| = q|n| \Rightarrow m \in n\mathbb{Z}$$

Meaning letting $m \in \ker\phi$ means $m \in n\mathbb{Z}$ making $\ker\phi \subseteq n\mathbb{Z}$, as requires.

Lemma: If there exists a non-zero $n \in \ker\phi$ then $\ker\phi = d\mathbb{Z}$ where d is the smallest divisor of n in $\ker\phi$

(Here the divisors are positive).

From second previous lemma there exists non-zero $m \in \ker\phi$ such that for nonzero $k \in \ker\phi \Rightarrow |k| \geq |m|$. From the next lemma we get $\ker\phi \subseteq m\mathbb{Z}$. From the first lemma we get $\ker\phi \supseteq m\mathbb{Z}$, combining them we get:

$$\ker\phi = m\mathbb{Z}$$

But nonzero $n \in \ker \phi = m\mathbb{Z}$ implies $m|n$.

But $k \in \ker \phi \Rightarrow |k| \geq |m|$ implies that all other divisors of n that are in $\ker \phi$ are greater than equal to m in magnitude, otherwise would cause a contradiction, hence m is the smallest divisor of n in $\ker \phi$, as required.

(Note that $d\mathbb{Z} \cong -d\mathbb{Z}$)

Lemma: If R is finite than there exits nonzero $n \in \ker \phi$

From the pidgin hole principle if we have $|R| + 1$ elements of $\mathbb{Z}$ than there are at least two distinct elements, m_0 and m_1, such that:

$$\phi(m_0) = \phi(m_1)$$

Hence:

$$\phi(m_0 - m_1) = 0$$

Since m_0 and m_1 are distinct $m_0 - m_1 \neq 0$ and hence $n = m_0 - m_1$ works.

Corollary: If R is finite then $\operatorname{img} \phi \cong \mathbb{Z}/n\mathbb{Z}$ for some nonzero n

Combine the previous theorem and lemma.

2.11.5 An Observation on Order

Lemma: $\operatorname{img} \phi \subseteq \langle 1_R \rangle$:

Because ϕ was defined by:

$$\phi(1) = 1_R, \quad \phi(n \pm 1) = \phi(n) \pm \phi(1_R) = \phi(n) \pm 1_R$$

The closure of $\langle 1_R \rangle$ means $\phi(n) \in \langle 1_R \rangle$ implies $\phi(n \pm 1)$ is. With the base case of $\phi(1) = 1_R \in \langle 1_R \rangle$ the lemma follows from induction.

Theorem: Let $\langle 1_R \rangle$ be finite. Then $\operatorname{img} \phi \cong \mathbb{Z}/|\langle 1_R \rangle|\mathbb{Z} \cong \langle 1_R \rangle$

By definition $\langle 1_R \rangle$ is the smallest subring of R containing 1_R, hence the smallest subring. From the first ring isomorphism theorem $\operatorname{img} \phi$ is a subring of R, hence:

$$|\operatorname{img} \phi| \geq |\langle 1_R \rangle|$$

But from the previous lemma we have:

$$|\operatorname{img}\phi| \leq |\langle 1_R \rangle|$$

The desired result follows from $\langle 1_R \rangle$ being cyclic.

Remark:

We expect $\langle a \rangle$ meaning "The smallest substructure containing a" to be well defined in any structure S that has closure. By closure I means that for some functions ϕ_n for all $U \subseteq S$ we have $\phi_n(U) \in S$. And by substructure I means some $R \subseteq S$ where closure still works for R.

Firstly, a is a member of the superstructure so theres is at least one possible set. Secondly, assume A and B be two minimal substructures. Now the closure of the substructures means $U \subseteq A \cap B$ implies $\phi_n(U) \subseteq A$ and $\phi_n(U) \subseteq B$ and hence $U \subseteq A \cap B \Rightarrow \phi_n(U) \in A \cap B$ making $A \cap B$ a substructure. Hence, from the minimality of A and B we have $A = A \cap B = B$.

I assume this argument can be made better with set/category theory.

Theorem: There is only one ring homomorphism from $\mathbb{Z}$ to a ring R

Given two homomorphisms ϕ_0 and ϕ_1 define:

$$d(n) = \phi_1(n) - \phi_0(n)$$

We have:

$$d(1) = \phi_1(1) - \phi_0(1) = 1_R - 1_R = 0$$

And:

$$\begin{aligned} d(n \pm 1) =& \phi_1(n \pm 1) - \phi_0(n \pm 1) \\ =& \phi_1(n) \pm 1_R - \phi_0(n) \mp 1_R \\ =& \phi_1(n) - \phi_0(n) \\ =& d(n) \end{aligned}$$

Hence, by induction, $d(n) = 0$ and $\phi_1 = \phi_0$.

Remark:

This observation isn't better or worse than the first observation, but it's result was what I wanted from the first, and it's more elegant, and it's better than the final corollary.

Either way, because of this uniqueness combined with $\operatorname{img}\phi \cong \langle 1_R \rangle|$ means we can identify $n_R = \phi(n)$ and have that notation work as implied.

2.11.6 Actual Subfield

I forgot to write about how the above results actually relates to finite subfields. This is just a quick note so I don't forget anything, it's not that rigorous and may include mistakes. (Like R has to be commutative for using polynomials to make sense.)

Theorem: $\langle a \rangle$ from polynomial substitution.

Let R be a ring and $s_a : \langle 1_R \rangle[X] \to R$ where:

$$s_a\left(\sum_n p_n X^n\right) = \sum_n p_n a^n$$

From the ring isomorphism theorem $\operatorname{img} s_a$ is a subring of R. Since $s_a(X) = a$ by the minimality of $\langle a \rangle$ we have:

$$\operatorname{img} s_a \supseteq \langle a \rangle$$

By closure of $\langle a \rangle$ and $s_a(X), s_a(1_R) \in R$ implies:

$$\operatorname{img} s_a \subseteq \langle a \rangle$$

Hence $\operatorname{img} s_a \cong \langle a \rangle$.

Remark:

You can naturally iterate on this with $s_{a,b,c} : \langle 1_R \rangle[X, Y, Z] \to R$ where $s_{a,b,c}[X] = a$,$s_{a,b,c}[Y] = b$, and $s_{a,b,c}[Z] = c$ to give:

$$\operatorname{img} s_{a,b,c} \cong \langle a, b, c \rangle$$

Condemned: I've condemned the rest of this section because it's confused, over complex, poorly phrase, and most importantly, wrong. I think I understood something at the time I wasn't able to capture and write down, but this isn't it either.

There is a new section for "Actual Subfields".

Corollary: If R is finite than there is a subset B such that B is a basis for R as a $\langle 1_R \rangle$-module.

Since R is finite we can have a $|R|$ variable polynomial which can interrelate the previous result as a $\langle 1_R \rangle$-module. Just prune off non-extending elements to get the basis.

Theorem: If R is a field then so it $\langle 1_R \rangle$.

Let $r \in \langle 1_R \rangle$ we have:

$$\begin{aligned} 1_R =& r \times \left(\sum_n p_n b_n \right) \\ 1_R =& \sum_n r p_n b_n \end{aligned}$$

Hence $rp_{n>0} = 0$, since all fields are domains ($ab = 0 \Rightarrow a = 0b^{-1}$) hence:

$$p_{n>0} = 0$$

Hence $\sum_n p_n b_n = p_0 \in \langle 1_R \rangle$ making it a field.

Corollary: If R is a field then $|\langle 1_R \rangle|$ is prime.

Recall that $\langle 1_R \rangle$ is cyclic and the only cyclic finite fields are the ones with prime order.

Notes

Just trying to write the previous process more formally, may not flesh out fully.

Let r_n be an enumeration of R, consider the functions:

$$\begin{aligned} s_n :\ & R_n[X] \longrightarrow R \\ & \sum_k p_k X^k \mapsto \sum_k p_k r_n^k \end{aligned}$$

Where $R_0 = \langle 1_R \rangle$ and $R_n = \operatorname{img} s_{n-1}$. We have:

$$R_0 \subseteq R_1 \subseteq R_2 \subseteq \cdots \subseteq R_{|R|} \cong R$$

The chain and set is finite and we can discard p_n where $R_n \cong R_{n-1}$ to get a set $B = \{p_{n'}\}$ where $r \in R$ implies r is a linear combination of powers of elements of B with coefficients in $\langle 1_R \rangle$.

2.11.7 Ordered Field

An ordered field is field with an additional relation $<$ such that for all x, y, $z \in \mathbb{F}$:

- Exactly one of $x < y$, $y < x$, or $x = y$ hold.
- If $x < y$ and $y < z$ implies $x < z$.
- If $x < y$ then $z + x < z + y$.
- If $0 < x$ and $0 < y$ then $0 < xy$.

(Other formalizations exist, but I will use these).

Interestingly the smallest subfield in every ordered field is isomorphic to $\mathbb{Q}$. This can be seen by defined a function $f : \mathbb{Q} \to \mathbb{F}$ such that:

$$f\left(\frac{p}{q}\right) = \phi(p)\phi(q)^{-1}$$

$\operatorname{img} f$ is the smallest subfield of f. For the same reasons ϕ is for rings, but includes reciprocals.

Now $0 < 1$ implies that $n < m \Rightarrow \phi(n) < \phi(m)$ by induction. Meaning $\operatorname{img} \phi \cong \mathbb{Z}$ since if $\phi(n) = \phi(m)$ we have, WLOG $n < m$, both $\phi(n-1) < \phi(m-1)$ and $\phi(n) = \phi(m) \Rightarrow \phi(n-1) = \phi(m-1)$, a contradiction.

But since:

$$f\left(\frac{1}{n}\right) f(n) = \phi(n)^{-1}\phi(n) = 1$$

Implies f is a field. Hence $\operatorname{img} f \cong \mathbb{Q}$ as required.

2.11.8 Actual Subfields #2

Theorem: If $\mathbb{F}$ is a field than all finite subring $R \subseteq \mathbb{F}$ are a fields.

There are actually a few cool ways to show this:

1: The most undergrad is pidgin-holing. Let $r \in R$, since R is finite there is at two $m < n \leq |R|$ such that:

$$r^n = r^m$$

Because r is a member of a field we can cancel $m - 1$ rs to give:

$$1 = r^{n-m-1}$$

If $n - m - 1 = 1$ then $r = 1$ and hence $r^{-1} = 1$. If $n - m - 1 > 1$ we have:

$$r^{n-m-2} = r^{-1}$$

2: For $r \in R$ let $f : R \to R$ such that:

$$f(x) = rx$$

f is injective since:

$$\begin{aligned} f(x) =& f(y) \\ rx =& ry \\ r(x - y) =& 0 \end{aligned}$$

All fields are domains so when $r \neq 0$ we have $x = y$.

Because f is a function between two finite groups of the same side it being injective means it's surjective. Hence there exits some x_0 such that:

$$1 = f(x_0) = rx_0 \Rightarrow r^{-1} = x_0$$

Corollary: If $\mathbb{F}$ is a field with finite $n = |\langle 1_{\mathbb{F}} \rangle|$ than n is prime.

Recall that $\langle 1_{\mathbb{F}} \rangle$ is cyclic and the only cyclic finite fields are the ones with prime order.

2.12 Dominated Convergence Theorem

Consider the following sequences of integrals:

$$I_n = \int_0^1 \log(t)^2 (1-t)^n \, dt$$

It's related to the Harmonic numbers, which is why it peaked my interest, but it was presented in context where we wished to understand its asymptotics.

The book answer of expressing it in terms of hypergeometric functions will be skipped, because it's boring.

2.12.1 Cauchy-Schwartz

$$\begin{aligned}\int_0^1 \log(t)^2 (1-t)^n \, dt \leq & \sqrt{\int_0^1 \log^4(t) \, dt \int_0^1 (1-t)^{2n} \, dt} \\ = & \sqrt{4! \frac{1}{2n+1}} \to 0\end{aligned}$$

Which by the noticing the sign of I_n and application of the sandwich theorem shows that $I_n \to 0$.

2.12.2 Dominated Convergence Theorem

As suggested by the section name, this is the result I really want to take notes about. If a sequence of functions f_n pointwise converge to f and are dominated by some integrable function g such that:

$$|f_n(t)| < g(t)$$

We have:

$$\int_S f_n \, d\mu \to \int_S f \, d\mu$$

Well rewrite the integral as:

$$\begin{aligned}\frac{n}{\log(n)^2}\int_0^1 \log(t)^2(1-t)^n\,dt &= \int_0^1 \left(\frac{(\log(nt)-\log(n)}{\log(n)}\right)^2 \left(1-\frac{nt}{n}\right)^n d\,nt \\ &= \int_0^n \left(\frac{\log(t)}{\log(n)}-1\right)^2 \left(1-\frac{t}{n}\right)^n dt \\ &= \int_0^\infty \chi_{[0,n]}(t)\left(1-\frac{\log(t)}{\log(n)}\right)^2 \left(1-\frac{t}{n}\right)^n dt\end{aligned}$$

Set:

$$f_n(t) = \overbrace{\chi_{[0,n]}(t)\left(1-\frac{\log(t)}{\log(n)}\right)^2}^{g_n(x)}\overbrace{\left(1-\frac{t}{n}\right)^n}^{h_n(t)}$$

We have (pointwise):

$$f_n(t) \to \exp(-t)$$

We also have $h_n(t) \leq \exp(-t)$ and $g_n(t) \leq (1-\log(t))^2$, meaning:

$$|f_n(t)| \leq (1-\log(t))^2 \exp(-t)$$

Giving the desired result (Assuming we can show the RHS converges, which I'm pretty sure we can since near 0 we have $\int(1-\log(t))^2\,dt = t(\log(t)^2 - 4\log(t)+5)$ and far away we have $\exp(-t)$).

$$\frac{n}{\log(n)^2}\int_0^\infty \log(t)^2(1-t)^n\,dt \to \int_0^\infty \exp(-t)\,dt = 1$$

2.13 Divisibility of $(p+1)(p-1)$

It is well known that if p is prime than either p is a divisor of 6 or has the remainder 1 or 5 when divided by 6. This can be verified by noting that a remainder of 0,2, or 4 would make p divisible by 2 and likewise for 0 and 3 for 3.

A cool corollary I saw today was that p is a divisor of 6 or 24 is a divisor of $(p+1)(p-1)$. This is also verified by cases:

$p \mod 12$	1	5	7	11
$\gcd(p+1, 12)$	2	6	4	12
$\gcd(p-1, 12)$	12	4	6	2

The converse isn't true. Since $24 \cdot 3 = 72$, but $(7-1)(7+1) = 48 < 72$ and $(11-1)(11+1) = 120 > 72$.

2.14 Matrix Function Evaluation using Cayley-Hamilton

(This is something I'm 70% sure I've seen before but I saw it again today). (Assume the field is $\mathbb{R}$, don't feel like dealing with others).

The Cayley-Hamilton theorem states that a matrix is the root of it's own characteristic equation. A cool application of this is simplifying the evaluation of (analytic) matrix functions.

Let M be a matrix with characteristic polynomial p, and eigenvalues λ_i with multiplicity m_i. Let the analytic function f be given by:

$$f(x) = \sum_{k=0}^{\infty} f_k x^k$$

Define a new function r that is interpolated on λ_i such that:

$$0 \leq k < m_i \Rightarrow \left.\frac{d^k f}{dx^k}\right|_{\lambda_i} = \left.\frac{d^k r}{dx^k}\right|_{\lambda_i}$$

This means that for some analytic function q we have:

$$f(x) = q(x)p(x) + r(x)$$

Now convert these analytic functions to analytic matrix functions in the natural way. From Cayley-Hamilton theorem we have:

$$f(M) = q(M)p(M) + r(M) = r(M)$$

2.14.1 Worked Example

Let M be a 2×2 matrix with two distinct eigenvalues. Let $f(x) = \exp(tx)$ then:

$$r(x) = (x - \lambda_0)\frac{\exp(t\lambda_1)}{\lambda_1 - \lambda_0} + (x - \lambda_1)\frac{\exp(t\lambda_0)}{\lambda_0 - \lambda_1}$$

Hence:

$$f(M) = r(M) = (M - \lambda_0 I)\frac{\exp(t\lambda_1)}{\lambda_1 - \lambda_0} + (M - \lambda_1 I)\frac{\exp(t\lambda_0)}{\lambda_0 - \lambda_1}$$

2.14.2 Original Application

The original problem was to minimize:

$$f(X) = \sum_{k=0}^{n} \left\| \exp\left(\frac{2\pi}{k} B\right) A - X \right\|^2$$

Through some procedure similar to the above reasoning it was obtained:

$$\exp(\alpha B) = I + \sin(\alpha) B + (1 - \cos(\alpha)) B^2$$

Hence:

$$\begin{aligned}
&\sum_{k=0}^{n} \left\| \exp\left(\frac{2\pi}{k} B\right) A - X \right\|^2 \\
=&\sum_{k=0}^{n} \left\| \left(I + \sin\left(\frac{2\pi}{k}\right) B + \left(1 - \cos\left(\frac{2\pi}{k}\right)\right) B^2 \right) A - X \right\|^2 \\
=&\sum_{k=0}^{n} \left\| \left((I + B^2)A - X \right) + \left(\sin\left(\frac{2\pi}{k}\right) B + \cos\left(\frac{2\pi}{k}\right) B^2 \right) A \right\|^2
\end{aligned}$$

The left term independent of k and the right term is independent of X. This means each term of the sum is minimized at:

$$X = (I + B^2)A$$

And since each term is minimized so is the whole sum.

2.14.3 Generalization of the Worked Example

Let M be a matrix with characteristic polynomial p, distinct eigenvalues λ_i, and corresponding eigenvectors v_i. The Lagrange functions $\ell_i(x)$ on $\{\lambda_i\}$ are given by:

$$\ell_i(x) = \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j}$$

And have the standard desirable property that:

$$\ell_i(x) = \begin{cases} 1 & x = \lambda_i \\ 0 & x = \lambda_j \text{ where } i \neq j \end{cases}$$

From the distinctness of λ_i:

$$p(x)|\ell_i(x)(x-\lambda_i)$$

(Where we don't necessarily have equality because of the denominators in $\ell_i(x)$). From Cayley-Hamilton theorem we have:

$$\ell_i(M)(M-\lambda_i I)=0 \Rightarrow \ell_i(M)M=\lambda_i\ell_i(M)$$

Which is pretty cool by itself, since ℓ_i is like an eigenmatrix, but also gives:

$$\begin{aligned}\ell_i(M)Mv_j =&\lambda_i\ell_i(M)v_j\\ \Rightarrow \lambda_j\ell_i(M)v_j =&\lambda_i\ell_i(M)v_j\end{aligned}$$

Hence either $\ell_i(M)v_j=0$ or $\lambda_i=\lambda_j$, but the eigenvalues are unique meaning $\ell_i(M)v_j=0$ or $i=j$.

This property is useful when we consider the standard remainder function r written as:

$$r(x)=\sum_i f(\lambda_i)\ell_i(x)$$

Giving:

$$\begin{aligned}f(M) =&\sum_i f(\lambda_i)\ell_i(M)\\ \Rightarrow f(M)\sum_j u_jv_j =&\sum_j u_jf(M)v_j\\ =&\sum_j u_j\sum_i f(\lambda_i)\ell_i(M)v_j\\ =&\sum_j u_jf(\lambda_j)\ell_j(M)v_j\end{aligned}$$

Hence the value of the product, over the space spanned by the eigenvalues of A, is a dot product of a term only dependent on the vector u and a term only dependent on the matrix and function.

An enlightening example of this is when $f(M)=\exp(tM)$, which has a natural interpretation as the continuous application of M for a period of t time, and gives an initial term, a time-dependent term, and a symmetry of M term:

$$\exp(tA)u=\sum_j u_j\exp(t\lambda_j)\ell_j(M)v_j$$

2.14.4 Derivatives Generalization:

I want to generalize the previous results for eigenvalues with multiplicities. (I just want to quickly write this down and come back latter if I feel like it, it has not been proof read) This will be with easier with a definition and two lemma:.

Let M be a matrix with characteristic polynomial p, eigenvalues λ_i, corresponding eigenvectors v_i and multiplicities m_i.

Modified Lagrange Function:

Define $\ell_i(x)$ as:

$$\ell_i(x) = \prod_{j \neq i} \left(\frac{x - \lambda_j}{\lambda_i - \lambda_j} \right)^{m_j}$$

We still have:

$$\ell_i(x) = \begin{cases} 1 & x = \lambda_i \\ 0 & x = \lambda_j \text{ where } i \neq j \end{cases}$$

But also for $n < m_j$ and $i \neq j$:

$$\ell_i^{(n)}(\lambda_j) = 0$$

this can be verified from the product rule.

Lemma:

From induction on n we have:

$$(M - \lambda_i I)^n v_j = (\lambda_i - \lambda_j)^n v_j$$

Recall that:

$$p(x) | \ell_i(x)(x - \lambda_i)^{m_i}$$

Hence:

$$\ell_i(M)(M - \lambda_i I)^{m_i} = 0$$

And:

$$(\lambda_j - \lambda_i)^{m_i} \ell_i(M) v_j = 0$$

Hence either $\ell_i(M)v_j = 0$ or $\lambda_i = \lambda_j$.

Lemma: Interpolation

Define:

$$r(x) = \sum_i \ell_i(x) \sum_{k=1}^{m_i} p_n \frac{(x-\lambda_i)^n}{n!}$$

Where p_n satisfy:

$$f^{(d)}(\lambda_i) = \sum_{k=0}^{d} p_k \ell_i^{(d-k)}(\lambda_i)$$

(Note that this is just a system of linear equations and can easily be solved through a linear equation).

Result:

$$f(M) = r(M) = \sum_i \ell_i(M) \sum_{k=1}^{m_i} p_n \frac{(M-\lambda_i I)^n}{n!}$$

In particular let $u = \sum_j u_j v_j$

$$\begin{aligned} f(M)u &= \sum_i \ell_i(M) \sum_{k=1}^{m_i} p_n \frac{(M-\lambda_i I)^n}{n!} u \\ &= \sum_i p_0 u_i \ell_i(M) v_i \end{aligned}$$

Since if $n > 0$ either $\ell_i(M)v_j = 0$ or $(M-\lambda_i I)^n v_j = 0$.

2.15 Simplicial Homology

I need to revise Simplicial Homology so I'm going to give some definitions and then work some examples.

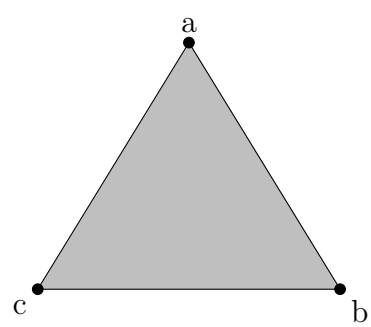
2.15.1 Simplicial Definitions

Simplex:

A simplex is a generalization of a triangle to higher/lower dimensions:

- 0-simplex is a point
- 1-simplex is a segment
- 2-simplex is a triangle
- 3-simplex is a tetrahedron

For our purposes we specify a simplex by with an ordered word written from the set of points. For example a is a point, bc is a segment, and abc is a triangle:



Two important remarks. Firstly, $abc \neq acb$ the former one goes around counter-clockwise and the later clockwise. Secondly, the set $\{ab, bc, bc\}$ is not a triangle and is not the same as abc or $\{abc\}$. It's all the edges of a triangle but not it's interior.

Face:

n 0-simplexes defines a n-simplex and is called the face of the points. Conversely the vertices that define a simplex and all the faces that come from that simplex are called the faces of the original simplex.

Geometric Simplicial Complex

A Geometric Simplicial Complex $\mathcal{K}$ is a set of simplexes such that:

1. Every face of an element of $\mathcal{K}$ is also an element of $\mathcal{K}$.
2. The non-empty intersection of two simplexes σ_0, $\sigma_1 \in \mathcal{K}$ is a face of both σ_1 and σ_0.

Chain

A k-chain of $\mathcal{K}$ is a formal linear sum of k-simplexes. A 2-chain is naturally interpreted as a, possible disconnected and repeated, path through $\mathcal{K}$.

More formally, given a complex $\mathcal{K}$ the group of n-chain $C_n(\mathcal{K})$ is a free $\mathbb{Z}$-module over the n-simplexes where two simplexes are equal if they have the same orientation[8] and additive inverse if opposite. (Define $C_{-1} = 0$ for convenience).

For example: $ab + bc$ is the path from a to c through b, $abc = -acb$ are the same triangle with different handedness, $2ab + dc$ is going over ab twice then jumping to dc.

Although the lower dimensions have intuitive geometric interpretation the whole configuration can more intuitively viewed as a combinatoric exercise.

[8]The words are even permutation of each other.

2.15.2 Homology Definitions

Boundary Operator

Given a simplex σ define σ_k as the same word a σ with the kth point removed. The boundary operator $\partial_k : C_k \to C_{k-1}$ is defined as:

$$\partial_k(\sigma) = \sum_{i=0}^{k} (-1)^i \sigma_i$$

The boundary of a 0-simplex is 0, the boundary of a 1-simplex is final point minus inital, the boundary of a 2-simplex is it's normal boundary (hence the name).

$$\begin{aligned} \partial_0(a) &= 0 \\ \partial_1(ab) &= b - a \\ \partial_2(abc) &= ac + cb + bc \end{aligned}$$

Cycles Subgroup

The cycles subgroup, Z_n, of C_n is the kernel of ∂_n:

$$Z_n = \ker \partial_n$$

The name comes from the $n = 1$ case where a $x \in \ker \partial_n$ iff the 1-chain is cyclic path.

Boundaries Subgroup

The boundaries subgroup, B_n, of C_n is the image of ∂_{n+1}:

$$B_n = \operatorname{img} \partial_{n+1}$$

The name comes from the $n = 1$ case where $\partial_2(x)$ is a 1-chain going over the boundary of x.

Homology Group

The boundary of a boundary is 0. For intuition consider the $n = 1$ case, the boundary is a cyclic path and hence the boundary of a boundary is zero.

In the general case the boundary of the boundary of σ is a sum of words of σ with two letters removed where each word appears twice, once for when the earlier letter is removed first then the last and again in the opposite order. Let $\sigma_{i,j}$ be the word obtained removing the ith and j^{th} letter.

When $i < j$ we have:

$$\sigma_{i,j} = (\sigma_i)_j$$

Since taking i^{th} out first doesn't effect removing j^{th}. And when taking the jth element first means the ith element is now one letter over:

$$\sigma_{i,j} = (\sigma_j)_{i-1}$$

Now observe that for the sum for the boundary of th boundary:

$$\begin{aligned}\partial_k(\partial_{k+1}(\sigma)) &= \sum_{i=0}^{k}(-1)^i \sum_{j=0}^{k+1}(-1)^j(\sigma_j)_i \\ &= \sum_{i=0}^{k}\sum_{j=0}^{k+1}(-1)^{i+j}(\sigma_j)_i\end{aligned}$$

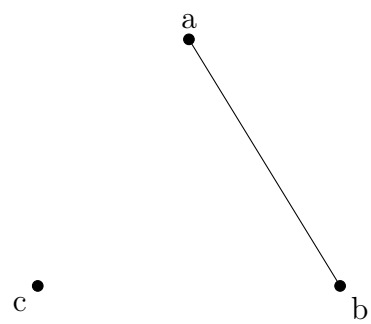
Meaning the $j-1$ in the index of the j first case changes the sign of $\sigma_{i,j}$ compared to the i first case. Hence the terms cancel and the sum is zero.

A direct consequence is that $\operatorname{img}\partial_{n+1}$ is a normal subgroup of $\ker\partial_n$ meaning we can use the fundamental isomorphism theorem to define the quotient group:

$$H_n(\mathcal{K}) = \frac{Z_n}{B_n} = \frac{\ker\partial_n}{\operatorname{img}\partial_{n+1}}$$

2.15.3 Worked Examples

Disconnected example



$$\mathcal{K} = \{a, b, c, ab\}$$

There's no simplexes of dimensions larger than 3 hence $C_{n\geq 3} = 0$ and $H_{n\geq 2}(\mathcal{K}) = 0$. Hence we have:

$$0 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

Where:

$$C_1 = \langle ab \rangle, \quad C_0 = \langle a, b, c \rangle$$

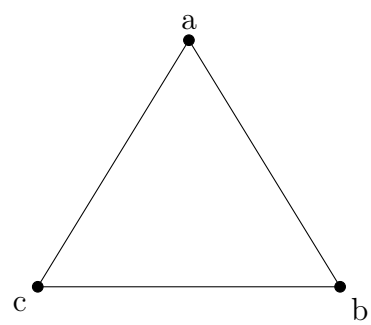
All results are pretty trivial:

$$\begin{aligned} \operatorname{img} \partial_2 &= 0 & \ker \partial_1 &= 0 \\ \operatorname{img} \partial_1 &= \langle a - b \rangle & \ker \partial_0 &= \langle a, b, c \rangle \end{aligned}$$

Gives the homologies:

$$\begin{aligned} H_0(\mathcal{K}) &= \frac{\ker \partial_0}{\operatorname{img} \partial_1} = \frac{\langle a, b, c \rangle}{\langle a - b \rangle} = \langle a, c \rangle = \mathbb{Z}^2 \\ H_1(\mathcal{K}) &= \frac{\ker \partial_1}{\operatorname{img} \partial_2} = 0 \end{aligned}$$

Triangle without Interior



$$\mathcal{K} = \{a, b, c, ab, ac, bc\}$$

Like with the previous example $H_{n\geq 2}(\mathcal{K}) = 0$. Hence we have:

$$0 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

Where:

$$C_1 = \langle ab, ac, bc \rangle, \quad C_0 = \langle a, b, c \rangle$$

The ker and img are given bellow:

$$\begin{aligned} \operatorname{img} \partial_2 &= 0 & \ker \partial_1 &= \langle ab + bc + ca \rangle \\ \operatorname{img} \partial_1 &= \langle a - b, a - c, b - c \rangle & \ker \partial_0 &= \langle a, b, c \rangle \end{aligned}$$

Proof for $\ker \partial_1$

$$\begin{aligned} &\partial_1(\alpha\, ab + \beta\, bc + \gamma\, ca) \\ =&\alpha(a - b) + \beta(b - c) + \gamma(c - a) \\ =&(\alpha - \gamma)a + (-\alpha + \beta)b + (-\beta + \gamma)c \end{aligned}$$

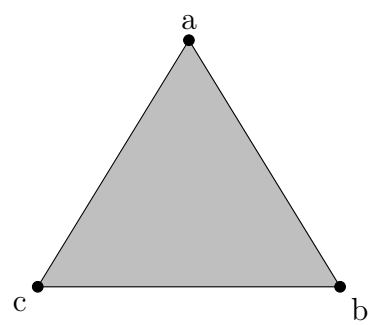
It might already be intuative that you can only choose one of α, β, γ then the rest are fixed. But you can also use the following module equation:

$$\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Gives the homologies:

$$\begin{aligned} H_0(\mathcal{K}) =& \frac{\ker \partial_0}{\operatorname{img} \partial_1} = \frac{\langle a, b, c \rangle}{\langle a - b, a - c, b - c \rangle} = \langle a \rangle = \mathbb{Z} \\ H_1(\mathcal{K}) =& \frac{\ker \partial_1}{\operatorname{img} \partial_2} = \frac{\langle ab + bc + ca \rangle}{0} = \langle ab + bc + ca \rangle = \mathbb{Z} \end{aligned}$$

Triangle without Interior



$$\mathcal{K} = \{a, b, c, ab, ac, bc, abc\}$$

There are no simplexes with order larger than 3 hence like in previous examples $H_{n\geq 3}(\mathcal{K}) = 0$. Hence we have:

$$0 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

Where:

$$C_2 = \langle abc \rangle, \quad C_1 = \langle ab, ac, bc \rangle, \quad C_0 = \langle a, b, c \rangle$$

The ker and img are given bellow:

$$\begin{aligned}
&\operatorname{img} \partial_3 = 0 && \ker \partial_2 = 0 \\
&\operatorname{img} \partial_2 = \langle ab + bc + ca \rangle && \ker \partial_1 = \langle ab + bc + ca \rangle \\
&\operatorname{img} \partial_1 = \langle a - b, a - c, b - c \rangle && \ker \partial_0 = \langle a, b, c \rangle
\end{aligned}$$

Gives the homologies:

$$\begin{aligned}
H_0(\mathcal{K}) =& \frac{\ker \partial_0}{\operatorname{img} \partial_1} = \frac{\langle a, b, c \rangle}{\langle a - b, a - c, b - c \rangle} = \langle a \rangle = \mathbb{Z} \\
H_1(\mathcal{K}) =& \frac{\ker \partial_1}{\operatorname{img} \partial_2} = \frac{\langle ab + bc + ca \rangle}{\langle ab + bc + ca \rangle} = 0 \\
H_2(\mathcal{K}) =& \frac{\ker \partial_2}{\operatorname{img} \partial_3} = \frac{0}{0} = 0
\end{aligned}$$

2.15.4 Interpretation of Simplicial Homology

Let us summarise the worked example results:

Complex	H_0	H_1	H_2	$H_{n\geq 3}$
Disconnected	$\mathbb{Z}^2$	0	0	0
Triangle, without interior	$\mathbb{Z}$	$\mathbb{Z}$	0	0
Triangle, with interior	$\mathbb{Z}$	0	0	0

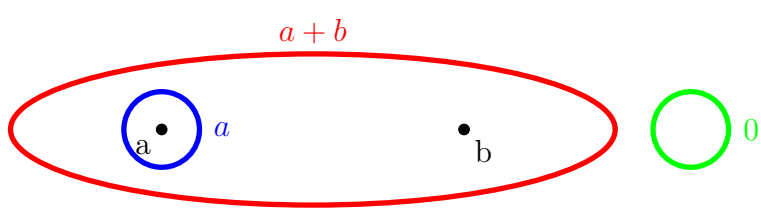
But what do they mean? In a nutshell the power of $\mathbb{Z}$ corresponds to a specific topological feature called an n-hole. A 0-hole is a connected part, a 1-hole is a regular hole, a 2-hole is a cavity. Hence the disconnected complex is made from two connected parts and the rest of one connected part. And the triangle without an interior has a hole and the others don't.

But why powers of $\mathbb{Z}$? And why does $\ker \partial_n / \operatorname{img} \partial_{n+1}$ give this result?

The rigorous answer has to do with homotopy, and the "Fundamental Homotopy Group" in particular, which is the math of drawing loops on things then pulling them and then seeing where they get stuck. Well worth a read but the basics are this:

For every hole you can wrap a loop around it multiple times and in both directions. Wrapping it around clockwise once is identified with 1, wrapping it around counter-clockwise twice is identified with -2, wrapping it around no hole is identified with 0 since when the loop is pulled taut nothing gets stuck.

If there are two hole a and b then the loops are identified with an element over the free $\mathbb{Z}$-module over $\{a, b\}$. For example $a + 2b$ a clockwise loop over a then two over b and you can add two elements by making a cut in both loops then gluing them together (making sure you preserve orientation).



The free $\mathbb{Z}$-module over $\{a, b\}$ is isomorphic to $\mathbb{Z}^2$ because you need two integers to explain all the loops you can make over two holes that stay when taut, which explains where the powers of $\mathbb{Z}$ came from.

$\ker \partial_n / \operatorname{img} \partial_{n+1}$ works because as noted in the definition of the chain group the elements of the chain group can jump around a bit. But the

requirement $\sigma \in \ker \partial_n$ makes it loop since it's boundary vanishes. And $\operatorname{img} \partial_{n+1}$ are all the loops that have a higher order faces enabling a taut loop to move freely. These steps combined give:

1. C_n contains all the loops but also completely broken path.
2. Use $\ker \partial_n$ to get all elements of C_n that are loops.
3. Modulo $\operatorname{img} \partial_{n+1}$ to remove all loops that don't exit when pulled taut.

Again, this explanation can be made more rigorous by studying homotopy.

2.16 Bounds of Three Series

Even though I'm sick I found an interesting problem on the internet. Given positive $a_1, b_1, c_1 > 0$ and:

$$\begin{aligned} a_{n+1} =& b_n + \frac{1}{c_n} \\ b_{n+1} =& c_n + \frac{1}{a_n} \\ c_{n+1} =& a_n + \frac{1}{b_n} \end{aligned}$$

Show that at least one of $a_{800}, b_{800}, c_{800}$ is greater than 40 and that all sequences are unbounded.

2.16.1 Lemma: Closed form for $x_{n+1} = x_n + \frac{k}{x_n}$

The closed form is this:

$$x_{n+1}^2 = x_1^2 + 2nk + k^2 \sum_{l=1}^{n} \frac{1}{x_l^2}$$

This result is nothing special, just a lot of algebra, the plan is to first get a result for the product of successive terms, then get perform induction on the product, and finally cleanup to get the final form.

$$\begin{aligned} x_{n+2}x_{n+1} &= \left(x_{n+1} + \frac{k}{x_{n+1}}\right) x_{n+1} \\ &= \left(x_n + \frac{k}{x_n} + \frac{k}{x_{n+1}}\right) x_{n+1} \\ &= x_{n+1}x_n + k + k\frac{x_{n+1}}{x_n} \\ &= x_{n+1}x_n + k + k\frac{1}{x_n}\left(x_n + \frac{k}{x_n}\right) \\ &= x_{n+1}x_n + 2k + \frac{k^2}{x_n^2} \end{aligned}$$

Induction gives:

$$x_{n+2}x_{n+1} = x_2x_1 + 2nk + k^2 \sum_{l=1}^{n} \frac{1}{x_l^2}$$

Then you clean up by using the initial relation on x_{n+2} and x_2.

2.16.2 Unbounded:

Let $s_n = a_n + b_n + c_n$, from the HM-AM inequality we have:

$$\frac{1}{a_n} + \frac{1}{b_n} + \frac{1}{c_n} \geq \frac{9}{s_n}$$

With equality if and only if $a_n = b_n = c_n$.

From the series definition we have:

$$s_{n+1} = s_n + \frac{1}{a_n} + \frac{1}{b_n} + \frac{1}{c_n} \geq s_n + \frac{9}{s_n}$$

Since $a_n = b_n = c_n \Rightarrow a_{n+1} = b_{n+1} = c_{n+1}$ we have s_n is bounded by:

$$s_{n+1}^2 \geq s_1^2 + 18n + 81 \sum_{l=1}^{n} \frac{1}{s_l^2}$$

Hence s_n grows at least as much as $3\sqrt{2n}$, which is unbounded.

If the sum is unbounded at least one of the series is unbounded. And if at least one series is unbounded the "swapping" nature of the series means that they all are.

2.16.3 800th terms:

Consider the bound sum:

$$s_{n+1}^2 \geq s_1^2 + 18n + 81 \sum_{l=1}^{n} \frac{1}{s_l^2}$$

If $s_n \leq 1$ then $s_{n+1} > 1$ hence at least half of the time $\frac{1}{s_l^2}$ must be greater than one. Giving the new bound:

$$s_{n+1}^2 > 18n + 81 \left\lfloor \frac{n}{2} \right\rfloor$$

(This isn't a particularly good bound, we could consider s_1 separately, but if works the result we need) Substituting in $n = 799$ gives:

$$s_{800}^2 > 18 \times 799 + 81 \times 398 > 18 \times 800 = 3^2 \times 40^2$$

Hence:

$$a_{800} + b_{800} + c_{800} = s_{800} > 3 \times 40$$

From pigeon holing at least one LHS term is greater than 40.

2.16.4 Remarks

This is yet another example of this Olympian style problem where I had to be prompted to consider the HM-AM inequality.

Although I was working with inequalities between elementary polynomial sums so wasn't that far away myself, and the other advice that was given was just straight up wrong, so I should probably just remember to use the inequality means in the future and let it go.

2.17 Summation by Parts

During a boring meeting I was reminded of a cool trick called "Summation by Parts". Given two sequences a_n and b_n define $B_n = \sum_{k=0}^{n} b_n$. Then we have:

$$\sum_{k=0}^{N} a_n b_n = a_N B_N - \sum_{n=0}^{N-1} B_n (a_{n+1} - a_n)$$

The name is clearly in reference to it being the discrete counterpart to integration by parts. With integration replaced with summation and derivatives with difference.

I will not be proving this formula, doing so is simple algebra.

2.17.1 Nicomachus's Theorem

Nicomachus's Theorem is the name of the following identity:

$$\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k \right)^2$$

It's a common object of recreational math and visual proofs in particular. And while I proofs of it have reached the point of annoyance, visual ones in particular, Summation of Parts provides another proof that motivates the squaring.

Let $\sigma_n = \sum_{k=0}^{n} k$ and set $a_n = \sigma_n$ and $b_n = n$. Hence $B_n = \sigma_n$ and $a_{n+1} - a_n = n + 1$ giving:

$$\sum_{k=0}^{N} n\sigma_n = \sigma_n^2 - \sum_{n=0}^{N-1} (n+1)\sigma_n$$

Since $\sigma_0 = 0$ we can shift the LHS sum up one to match the RHS's bound:

$$\sum_{k=0}^{N} n\sigma_n = \sigma_n^2 - \sum_{n=0}^{N} n\sigma_{n-1}$$

Giving:

$$\sum_{k=0}^{N} n(\sigma_n + \sigma_{n-1}) = \sigma_n^2$$

The observation that $\sigma_n + \sigma_{n-1} = n^2$ completes the proof.

It's worth nothing this proof doesn't require an explicit closed formula for σ_n. Instead only needing $\sigma_{n+1} - \sigma_n = n + 1$,$\sigma_{n+1} + \sigma_n = (n+1)^2$ and $\sigma_0 = 0$. Yes, a close form follows easily from these properties as:

$$\sigma_{n+1} = \frac{1}{2}((\sigma_{n+1} + \sigma_n) + (\sigma_{n+1} - \sigma_n))$$

But it's still a cool proof

2.17.2 Continuous Summation by Parts

Consider the case here a_n is given by a continuously differential function $a(x)$, that is $a_n = a(n)$. Likewise modify B_n into a continuous function, $B(x) = B_{\lfloor x \rfloor}$. Hence we can write the RHS summation as:

$$\begin{aligned}\sum_{n=0}^{N-1} B_n(a_{n+1} - a_n) &= \sum_{n=0}^{N-1} B(n)(a(n+1) - a(n)) \\ &= \sum_{n=0}^{N-1} B(n) \int_n^{n+1} a'(t)\, dt\end{aligned}$$

Note that $B(t)$ is constant for $t \in (n, n+1)$ allowing us to move it into the integral:

$$\begin{aligned}\sum_{n=0}^{N-1} B_n(a_{n+1} - a_n) &= \sum_{n=0}^{N-1} \int_n^{n+1} B(t)a'(t)\, dt \\ &= \int_0^n B(t)a'(t)\, dt\end{aligned}$$

Substitution into the original equation gives:

$$\sum_{k=0}^{N} a_n b_n = a_N B_N - \int_0^n B(t)a'(t)\, dt$$

If we cancel the fractional parts of the product and integral on the RHS we get the more general:

$$\sum_{0 \le n \le x} a_n b_n = a(x)B(x) - \int_0^x B(t)a'(t)\,dt$$

This isn't a particularly good formula for relating a sum to an integral, see the Euler-Maclaurin formula, but it's still good enough for some results:

Results:

Let $a(x) = \ln(x)$ and $b_n = 1$ then we have:

$$\sum_{1 \le n \le x} \ln(n) = \lfloor x \rfloor \ln(x) - \int_1^x t\,\frac{1}{t}\,dt = x\ln(x) - x + O(\ln(x))$$

Let $a(x) = \frac{1}{x}$ and $b_n = 1$

$$\sum_{1 \le n \le x} \frac{1}{n} = \frac{\lfloor x \rfloor}{x} - \int_1^x t\,\frac{-1}{t^2}\,dt = \ln(x) + O(1)$$

Both of these bounds are useful for computer science.

2.18 Rational Residue

I saw a cool result. That if P and Q are polynomials and Q has a first order root at a then the then:

$$\operatorname{Res}\left(\frac{P}{Q}, a\right) = \frac{P(a)}{Q'(a)}$$

Which can save a lot of time over factorizing, which is what I would have done in the past.

This follows directly from:

$$Q'(a) = \frac{d}{dz}(z-a)q(z)\bigg|_a = q(a)$$

and

$$\oint_\gamma \frac{P(z)}{Q(z)}\,dz = \oint_\gamma \frac{P(z)}{(z-a)q(z)}\,dz = 2\pi i\frac{P(a)}{q(a)}$$

I thought this might be generalizable since the general Leibniz rule on $Q(z) = (z-a)^2 q(z)$ gives:

$$Q''(a) = 2q(a), \quad Q'''(a) = 6q'(a)$$

Hence for a second order root:

$$\begin{aligned}\oint_\gamma \frac{P(z)}{Q(z)}\,dz = \oint_\gamma \frac{P(z)}{(z-a)^2 q(z)}\,dz \\ =&\pi i\frac{P'(a)q(a) - P(a)q'(a)}{q(a)^2} \\ =&2\pi i\frac{3P'(a)Q''(a) - P(a)Q'''(a)}{3Q'(a)^2}\end{aligned}$$

Giving:

$$\operatorname{Res}\left(\frac{P}{Q}, a\right) = \frac{3P'(a)Q''(a) - P(a)Q'''(a)}{3Q'(a)^2}$$

Which isn't as cool as the first order case, and I don't feel like dealing with higher derivatives.

Still cool though.

2.18.1 Actual closed form

There actually is a way to get a closed form without dealing with higher order derivatives of reciprocals. The idea is to use the Bézout trick to get a polynomial that acts like a reciprocal at $z = a$.

Let Q be a polynomial with an n ordered root at a. Then define q as:

$$Q(z) = (z-a)^n q(z)$$

Since $\gcd(q(z), (z-a)^n) = 1$ from the Bézout's identity, there exits polynomials r and s such that:

$$r(z)(z-a)^n + s(z)q(z) = 1$$

Hence:

$$\frac{P(z)}{(z-a)^n q(z)}(1 - s(z)q(z)) = \frac{P(z)}{q(z)} r(z)$$

Has no poles at a and vanishes in the contour integral. Meaning we can split up its terms to give:

$$\oint_\gamma \frac{P(z)}{(z-a)^n q(z)}\, dz = \oint_\gamma \frac{P(z)s(z)}{(z-a)^n}\, dz$$

Resulting in the clean:

$$\operatorname{Res}\left(\frac{P}{Q}, a\right) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} P(z)s(z)\bigg|_{z=a}$$

Remarks

You can use use the standard extended Euclid algorithm to calculate s from q, or q from s. But the relation:

$$r(z)(z-a)^n + s(z)q(z) = 1$$

Can be derived $n-1$ and set $z = a$ to get a linear system between derivatives of s and q. So why bother with Bézout at all? Well you would need to prove that the coefficients from the system converge to an analytic function, Bézout side steps this.

I'm also wondering when $s^{(n)}(a) = 0$, because we loose information about F in the sum when this happens. Also working backwards, can we have any s. Not $s(z) = z - a$ obviously, but we can add another power to keep it? $s(z) = z^2 + z - a$?

2.18.2 A Generalization

Generate new solutions to the Bézout Identity in the standard way:

$$[r(z) - h(z)s(z)](z-a)^n + s(z)[q(z) + (z-a)^n h(z)] = 1$$

If we choose h such that $r(z) - h(z)s(z)$ is holomorphic, easy since r and s are polynomials, we have, given γ is small enough to not add new poles from h:

$$\oint_\gamma \frac{P(z)}{(z-a)^n[q(z) + (z-a)^n h(z)]}\, dz = \oint_\gamma \frac{P(z)s(z)}{(z-a)^n}\, dz = \frac{1}{(n-1)!}\frac{d^{n-1}}{dz^{n-1}} P(z)s(z)\bigg|_{z=a}$$

Observe that all function $q'(z)$ such $(z-a)^n | q(z) - q'(z)$ can be written as $q'(z) = q(z) + (z-a)^n h(z)$ for some holomorphic h giving the pretty cool result that only the fist n derivatives of q' effect the value of the contour integral. (Also observe that $r'(z) = r(z) - h(z)s(z)$ means r' agrees with r on s helping motivate the LHS)

$q'(z) = 1$ gives $s(z) = 1 - (z-a)^n$, $r'(z) = 1$ and hence Cauchy's integral formula and hence generalizes it.

This could also be used to extract terms from the reciprocal of a generating function.

2.19 Weierstrass-Erdmann Corner Conditions

I saw an enlightening Calculus of Variation problem that involved some of the more interesting results from the field.

Find $x(t)$ such that:

$$I = \int_0^2 (1-\dot{x})^2 \dot{x}^2 \, dt$$

Is minimized with endpoints $x(0) = 0$ and $x(2) = 1$.

By inspection $F(t, x, \dot{x}) = (1-\dot{x})^2\dot{x}^2 \geq 0$ making $I \geq 0$ with the F vanishing iff $\dot{x} = 0, 1$. Hence any piecewise combinations of lines with gradients 0 or 1 from $(0,0)$ to $(2,1)$ works. For example, the solution of a straight line from $(0,0)$ to $(1,1)$ then to $(2,1)$ is a minimum. This solution is continuous but not differentiable.

Now lets use more standard methods. First calculate a table of useful properties of F:

$$\begin{aligned}
F_x =& 0 \\
F_{\dot{x}} =& 2\dot{x}(1-\dot{x})(1-2\dot{x}) \\
F_{\dot{x}^2} =& 12\dot{x}^2 - 12\dot{x} + 2 \\
\frac{d}{dt} F_{\dot{x}} =& \ddot{x} F_{\dot{x}^2} \\
=& \ddot{x}(12\dot{x}^2 - 12\dot{x} + 2)
\end{aligned}$$

The Euler-Lagrange equation, combined with the first and last line, means that all extremals take the form:

$$x(t) = at + b$$

These are lines, but a line directly between the endpoints is:

$$x(t) = \frac{1}{2}t$$

Meaning:

$$F = \frac{1}{16} > 0$$

This is clearly greater than the piecewise solutions, but lets pretend we don't know that and use some other results.

2.19.1 Only solution is not a Minimum

The Legendre-Clebsch condition says that in order for $x_0(t)$ to minimize $\int_U F(t, x, \dot{x})\, dt$ we require:

$$t \in U \Rightarrow F_{\dot{x}^2}(t, x_0, \dot{x_0}) \geq 0$$

But we have:

$$x(t) = \frac{1}{2}t \Rightarrow F_{\dot{x}^2} = -1$$

Meaning the only solution of the Euler-Lagrange equation isn't a minimum. Does that mean there are none? No, just none in the set of functions that match the Euler-Lagrange equations assumptions. (At least continuously differentiable, but the set may be bigger). But clearly minimal solutions exits from the beginning.

2.19.2 Corners

Continue to pretend we don't know the piecewise solutions. The Weierstrass-Erdmann corner condition states that for first-order derivate discontinuities, or "corners", at t_0 the functions on either side, x_0 and x_1 must satisfy:

$$F_{\dot{x}}(t_0, x_0, \dot{x_0}) = F_{\dot{x}}(t_0, x_1, \dot{x_1})$$

and

$$F_{\dot{x}}(t_0, x_0, \dot{x_0})\dot{x_0}(t_0) - F(t_0, x_0, \dot{x_0}) = F_{\dot{x}}(t_0, x_1, \dot{x_1})\dot{x_1}(t_0) - F(t_0, x_1, \dot{x_1})$$

Our F is pretty simple, only put a restrictions on $\dot{x_n}$,

$$\begin{aligned} \dot{x_0}(1 - \dot{x_0})(1 - 2\dot{x_0}) =& \dot{x_1}(1 - \dot{x_1})(1 - 2\dot{x_1}) \\ \dot{x_0}^2(1 - \dot{x_0})(1 - 3\dot{x_0}) =& \dot{x_1}^2(1 - \dot{x_1})(1 - 3\dot{x_1}) \end{aligned}$$

This means $\dot{x_0}, \dot{x_1} \in \{0, 1\}$. It's hard to see this from algebra alone, or maybe it's just that I'm tired, but plotting $x(1 - x)(1 - 2x)$ and $x^2(1 - x)(1 - 3x)$ shows it pretty well.

Observe that this is just the piecewise solutions we got from first inspection. Pretty cool.

2.19.3 Next Steps

There were other goodies in the place I found this problem.

First there are some notes here for implicit boundaries:

$$S(x, y) = 0$$

Which might be useful for shader stuff, but haven't pieced it together yet.

There is also an iterative method of solving Calculus of Variation problems. It might be interesting to inspect this iteration and see the differential solutions converge to the piecewise solutions. Or maybe something else cooler!!

2.20 Legendre Transform

I've used the Legendre transform a lot and always understood that it transformed between conjugate variables but never understood the motivation of its formal definition. Well I've thought of a good motivated argument.

Let f be a strictly convex differentiable function in x. Then $\frac{\partial f}{\partial x}$ is a strictly increasing function. This means $\frac{\partial f}{\partial x}$ is injective and we can defined a function g such that:

$$g\left(\left.\frac{\partial f}{\partial x}\right|_{x_0}\right) = x_0$$

And we can further define a new function f^*, up to an additive constant, such that:

$$\frac{\partial f^*}{\partial p} = g(p)$$

This is the Legendre Transform.

This transforms usefulness comes from looking at the functions differential. Let $f(x, y)$ and consider a point (x_0, y_0) where:

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = p, \quad \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = v$$

Giving the differential:

$$df = pdx + vdy$$

Now consider a $f^*(p, y)$ where f^*'s first argument is Legendre transformed from f's, and the second is unchanged. This gives:

$$df^* = xdp + vdy$$

You can see that the coefficient and differential of the first term has swapped, pretty useful.

This argument seems more motivated to me, but the domain of the transform can be widened to merely require f being convex on D by defining:

$$f^*(p) = \sup_{x \in D} \left[px - f(x)\right]$$

To see that this definition extends the strictly convex differentiable case derive the inner function by x and set to zero:

$$p - f'(x) = 0$$

But f is injective, so we can defined g as before and obtain:

$$f'(g(p)) = p$$

meaning the maximal value of x is $g(p)$, giving:

$$f^*(p) = pg(p) - f(g(p))$$

Deriving by p gives:

$$\frac{d}{dp} f^*(p) = g(p) + pg'(p) - g'(p)f'(g(p))$$

Substituting in $f'(g(p)) = p$ gives:

$$\frac{d}{dp} f^*(p) = g(p) + pg'(p) - g'(p)p = g(p)$$

As expected.

2.21 Information Theory based Entropy

I want to do some entropy/temperature calculations only using motivations from a information perspective. We will start with the classic problem:

$$\text{Maximize: } H = -\sum_i p_i \ln(p_i)$$
$$\text{Subject to: } \sum_i p_i = 1, \quad \sum_i p_i w_i = \mu$$

Start with a standard Lagrangian:

$$L = -\sum_i p_i \ln(p_i) - a\left(\sum_i p_i - 1\right) - b\left(\sum_i p_i w_i - \mu\right)$$

Giving:

$$p_i = \exp(-1 - a - bw_i)$$

We can easily remove a by normalizing:

$$p_i = \frac{1}{Z(b)} \exp(-bw_i)$$

Where:

$$Z(t) = \sum_i \exp(-tw_i)$$

Likewise for μ:

$$\mu = \sum_i p_i w_i = \frac{1}{Z(b)} \sum_i w_i \exp(-bw_i)$$

Substituting back into H:

$$\begin{aligned} H =& \frac{1}{Z(b)} \sum_i \left(\exp(-bw_i)(bw_i + \ln(Z(b))) \right) \\ =& \frac{1}{Z(b)} \left(bZ(b)\mu + Z(b)\ln(Z(b)) \right) \\ =& b\mu + \ln(Z(b)) \end{aligned}$$

The $b\mu$ term already lets us interpret b^{-1} as acting like temperature. As we'd expect an increase of μ to allows us to increase entropy because it allows

us to distribute more probability to larger w_i.

Let go further by noticing that:

$$\mu = -\frac{Z'(b)}{Z(b)} = -\frac{d}{d\,b}\ln(Z(b))$$

This lets us formalizes the statement "more probability going to larger w_i" from earlier.

$$\begin{aligned}\frac{dp_i}{d\mu} =& \left(\frac{-w_i}{Z(b)}\exp(-bw_i) - \frac{Z'(b)}{Z^2(b)}\exp(-bw_i)\right)\frac{d\,b}{d\,\mu}\\ =& p_i(\mu - w_i)\frac{d\,b}{d\,\mu}\end{aligned}$$

And also:

$$\frac{d\,H}{d\,b} = b\frac{d\,\mu}{d\,b} + \mu + \frac{Z'(b)}{Z(b)} = b\frac{d\,\mu}{d\,b} + \mu - \mu = b\frac{d\,\mu}{d\,b}$$

Which gives:

$$\frac{d\,H}{d\,\mu} = b$$

As expected.

2.21.1 A General Case

There's actually a lot of important probability distributions what have a similar form. The numerator is an exponential of a weight times a constant and the denominator is, naturally, the sum of these exponentials.

Start with a similar Lagrangian as before:

$$L = -\sum_i p_i \ln(p_i) - a\left(\sum_i p_i - 1\right) - bF$$

Where F is a function of just p_i. The goal is to extremize entropy while keeping normalization and the constraint F, giving:

$$p_i = \exp\left(-1 - a - b\frac{\partial F}{\partial p_i}\right)$$

Defining Z similar to before:

$$Z(t) = \sum_i \exp\left(-t\frac{\partial F}{\partial p_i}\right)$$

Lets us normalize out a like before:

$$p_i = \frac{1}{Z(b)} \exp\left(-b\frac{\partial F}{\partial p_i}\right)$$

We can also similarly define the average of the weights:

$$\begin{aligned}\sum_i p_i \frac{\partial F}{\partial p_i} &= \sum_i \frac{1}{Z(b)} \exp\left(-b\frac{\partial F}{\partial p_i}\right) \frac{\partial F}{\partial p_i} \\ &= -\frac{Z'(b)}{Z(b)} \\ &= -\frac{d}{d\,b} \ln(Z(b))\end{aligned}$$

2.22 Hadamard Matrix

I came across this cute set of matrices today:

$$H_{2^k} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes H_{2^{k-1}}, \quad H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

They are called the Hadamard matrices and their first interesting property is that their columns are orthogonal.

To see this consider two matrices A and B, we have:

$$(AB^T)_{i,j} = \sum_{k=1}^{n} A_{i,k} B_{j,k}$$

Which is the dot product of the ith and j^{th} columns of A and B. Hence H_n's columns being orthogonal is the same as:

$$(H_n H_n^T)_{i,j} = f(i)\delta_{i,j}$$

For induction assume:

$$H_{2^{k-1}} H_{2^{k-1}}^T = 2^{k-1} I_{2^{k-1}}$$

Hence:

$$\begin{aligned} H_{2^k} H_{2^k}^T &= \begin{bmatrix} H_{2^{k-1}} & H_{2^{k-1}} \\ H_{2^{k-1}} & -H_{2^{k-1}} \end{bmatrix} \begin{bmatrix} H_{2^{k-1}}^T & H_{2^{k-1}}^T \\ H_{2^{k-1}}^T & -H_{2^{k-1}}^T \end{bmatrix} \\ &= \begin{bmatrix} 2H_{2^{k-1}} H_{2^{k-1}}^T & H_{2^{k-1}} H_{2^{k-1}}^T - H_{2^{k-1}} H_{2^{k-1}}^T \\ H_{2^{k-1}} H_{2^{k-1}}^T - H_{2^{k-1}} H_{2^{k-1}}^T & 2H_{2^{k-1}} H_{2^{k-1}}^T \end{bmatrix} \\ &= \begin{bmatrix} 2^k I_{2^{k-1}} & 0 \\ 0 & 2^k I_{2^{k-1}} \end{bmatrix} \\ &= 2^k I_{2^k} \end{aligned}$$

Which shows H_n's columns are orthogonal, and also that:

$$H_n^{-1} = \frac{1}{2^n} H_n^T$$

Taking the determinates of both sides:

$$\det(H_n)^2 = \det(nI_n) = n^n$$

Hence:

$$|\det(H_n)| = n^{\frac{n}{2}}$$

And it actually turn out that all other $n \times n$ real matrices with elements of magnitude less then or equal to one have a determinate whose magnitude is less than this, which is pretty cool.

2.23 Hermitian Operators

This is some quick revision for something else.

For this section let A and B be hermitian operators in some complex vector space. If they have them, let a be an eigenvector of A with value α. Likewise for b and β for B.

2.23.1 Immediate Results

Theorem: All eigenvalues are real

$$\begin{aligned}\alpha\langle a, a\rangle =&\langle a, \alpha a\rangle\\ =&\langle a, Aa\rangle\\ =&\langle Aa, a\rangle\\ =&\bar{\alpha}\langle a, a\rangle\end{aligned}$$

Since eigenvectors can't be zero we require $\alpha = \bar{\alpha}$.

Theorem: $\ker A = \operatorname{img} A^{\perp}$

Assume $Ax = 0$ and let y be any arbitrary vector, then:

$$0 = \langle Ax, y\rangle = \langle x, Ay\rangle$$

Hence $x \in \operatorname{img} A^{\perp}$

2.23.2 Finite, Nonzero, Complex Vector Space

For this section we assume that A and B belong to a finite, nonzero, complex vector space.

A discussion on all these constraints will likely be included at the end. But the reason of the constraints is the following powerful theorem:

Theorem: If A is a linear operator on a finite, nonzero, complex vector space then there exits at least one unit eigenvector.

By the fundamental theorem of algebra the characteristic polynomial of A has at least one solution and hence A has at least one eigenvector.

Without loss of generality we can rescale the eigenvector to make it a unit vector.

Remark:

This existence theorem is very powerful and but uses all of the conditions. Finite to get a polynomial, nonzero so it's not a constant polynomial, complex to make that polynomial have a root.

Corollary: Let A be the transform of a nonzero invariant subspace V, then V includes a unit eigenvector of A:

V inherits finiteness and complex from its super space, we can also treat A as a linear operator on V by invariance, and since V is also assumed to be nonzero the previous theorem applies.

Lemma: Let E be a set of eigenvectors for A then A preserves $\operatorname{span}(E)^{\perp}$:

If $x \in \operatorname{span} E^{\perp}$ then for all c_n we have:

$$\begin{aligned}\langle Ax, \sum_n c_n a_n \rangle =& \langle x, A \sum_n c_n a_n \rangle \\ =& \langle x, \sum_n c_n \alpha_n a_n \rangle \\ =& 0\end{aligned}$$

Theorem: There exits an orthonormal basis of eigenvectors of A

From the existence theorem there exits a unit eigenvector a_0. Let $S_0 = \operatorname{span}(a_0)^{\perp}$, then by the previous lemma S_0 is an invariant subspace. Hence by the existence theorem there is a unit eigenvector $a_1 \in S_0$. Likewise define $S_1 = \operatorname{span}(\{a_0, a_1\})^{\perp}$ and so on until we get a basis.

Theorem: If A and B commute then there exits a mutuality orthogonal eigenvectors basis

From the existence theorem there is an eigenvector a of A. Define the set:

$$W = \operatorname{span}(\{B^n a \mid n \in \mathbb{N}\})$$

Let $w \in W$, then by definition we have $Bw \in W$ hence by the existence corollary there exits unit eigenvector vector w of B in W. From commutativity and the definition of W we also have $Aw = aw$. Hence w is an eigenvector of both operations. We can build up a basis with induction, like the previous theorem.

2.23.3 All Those Conditions

The critical points in the previous proofs was the existence theorem and induction on the eigenvectors. But the existence theorem didn't use the fact the operators where Hermitian, so surely we can remove some of those conditions!

Well we could weaken Complex to algebraically closed field without hermitian but we can't remove finite and nonzero. The nonzero condition isn't much of a concern, the vector field $\{0\}$ isn't interesting but to see why we can't remove finite consider the set of square-integrable functions on $[0, 1]$.

The operator $[Af](t) \mapsto tf(t)$ is clearly Hermitian because the integrating veritable is real:

$$\langle f, tg \rangle = \int_0^1 \overline{g(t)} t f(t)\, dt = \int_0^1 \overline{tg(t)} f(t)\, dt = \langle tf, g \rangle$$

But has no eigenvector as:

$$tf(t) = \alpha f(t)$$

Means $f(t)$ can only be nonzero at $t = \alpha$, which means only $f(\alpha)$ can be nonzero. Remembering that the vector space only cares about functions up to the equivalence class of being equal "almost everywhere" this function is the constant 0 function.

This operator is clearly important for functional analysis, so we can't define a set of nicer functions, so what do we do? There is one interesting result:

There exits a number λ equal to $||A||$ or $-||A||$ and a sequence of unitary vectors x_n such that:

$$\lim_{n \to \infty} \big(Ax_n - \lambda x_n \big) = 0$$

Which is like λ being an eigenvalue in the limit.

Applying this eigen-limit idea to square integral functions and we get the $\delta(t-\alpha)$ functions, which can be approached by many different sequence of limit functions.

2.24 Time Evolution of a Two Dimensional Quantum System

I was curious about the evolution of two dimensional quantum systems and decided to do a refresher.

Like all time-dependent problems start with the Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}\Psi = H\Psi$$

If Ψ is an eigenvector of H then we have:

$$i\hbar\frac{\partial}{\partial t}\Psi = E\Psi \Rightarrow \Psi = \exp(-i\hbar Et)\Psi_0$$

A two-dimensional system where the energy levels are the same has a trivial evolution, both gain phase at the same rate resulting in no observable change, so assume there are two eigenvalues $\mu \pm \delta$.

Let use working the orthonormal base of H and define:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad E = \begin{bmatrix} \mu - \delta & 0 \\ 0 & \mu + \delta \end{bmatrix} = \mu I - \delta Z$$

Then we have:

$$\Psi = \exp(-i\hbar Et)\Psi_0 = \exp(-i\hbar\mu t)\exp(i\hbar\delta Zt)\Psi_0$$

We will ignore the first factor as it's only relevant if working with a super system.

Consider the function $f(x) = \exp(i\hbar\delta tx)$ and apply the Cayley–Hamilton theorem to obtain:

$$\begin{aligned}\Psi &= \left(\exp(-i\hbar\delta t)\frac{Z+I}{2} - \exp(i\hbar\delta t)\frac{Z-I}{2}\right)\Psi_0 \\ &= \big[\cos(\hbar\delta t)I - i\sin(\hbar\delta t)Z\big]\Psi_0\end{aligned}$$

Observer that when $\delta t \to 0$ we have $\Psi \to \Psi_0$. This has two obvious sanity checks. That as we return to the beginning, $t \to 0$, the state returns to Φ_0

regardless of δ. And as the difference of the energy level shrinks, $\delta \to 0$, the state returns to Φ_0 regardless of t.

Other results include:

That for small δ we can keep the system in the Ψ_0 state for an inversely proportional time

That the system returns to Ψ_0 with a period of $\frac{\pi}{\hbar\delta}$. Notice that the period is halved from the naïve value, as we only require the sin term to vanish.

2.24.1 Change of Basis

Assume the Ψ_0 isn't given in the orthonormal base of H. Then we will need a change of basis matrix P and have:

$$\Psi = P^{-1}\exp(-i\hbar Et)P\Psi_0$$

We can just substitute our worked expression for $\exp(-i\hbar Et)$ to give

$$\Psi = \left[\cos(\hbar\delta t)I - i\sin(\hbar\delta t)P^{-1}ZP\right]\Psi_0$$

But lets pretend we didn't have the worked form, what would we do?

Well, interestingly we have:

$$P^{-1}\exp(-i\hbar Et)P\Psi_0 = \exp(-P^{-1}i\hbar EtP)\Psi_0$$

So things don't change much yet. And when it comes to applying the Cayley-Hamilton theorem we need the characteristic polynomials:

$$\begin{aligned} p_{P^{-1}MP}(x) &= \det(P^{-1}MP - xI) \\ &= \det(P^{-1}(M - xI)P) \\ &= \det(P^{-1})\det(M - xI)\det(P) \\ &= \det(M - xI) \end{aligned}$$

Meaning the roots, eigenvalues, of one are the roots of another. Meaning the result of Cayley-Hamilton is the same but with a different matrix subbed in.

From this point it's just algebra and left to the reader. But what's interesting is that all the operations involved worked so well with:

$$M \mapsto P^{-1}MP$$

Which might be physically motivated.

In abstract algebra we would call this type of map a "conjugation", but that name already has a meaning in linear algebra, so maybe "inner-automorphism" would be better.

2.25 Parabola

For a while I've wanted to know more about conic sections and have decided that the parabola is a good plaything to start with.

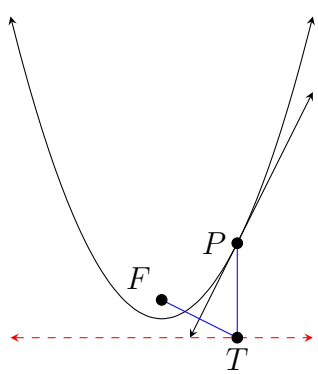
2.25.1 Analytic versus Synthetic

The analytic-synthetic distinction exists in multiple fields. Analysis means breaking something down into parts while synthesis builds something up.

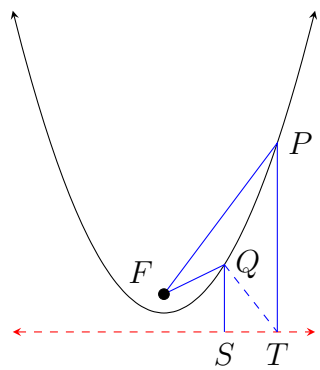
A classic example is analytic-synthetic truths from Kant. An analytic truth is ones whose truth comes from being broken down to definitions, "A number is even or it is odd" and "All bachelors are unmarried". A synthetic truth is one whose truth depends on the real world (or is "built" from it), "The sky is blue".

Like with all philosophy many will argue over my choice of examples, but in geometry the divide is less likely to start an argument. An analytic geometric argument breaks the figure down to coordinates while a synthetic one builds new figures up.

To demonstrate the difference I will present two argument for the same result: "Let a point P be on a parabola with focus F. Drop a line PT such that it is perpendicular to, and T is on, the directrix. Then the tangent going though P is the perpendicular bisector of FT."



Synthetic



Let F be the focus of the parabola, let P and Q be distinct points on the parabola, and let T and S be points of the directrix such that PT and QS are perpendicular to the directrix.

By definition of a parabola $|QF| = |QS|$ and $|PF| = |PT|$, and by construction $\triangle QST$ is a right angled triangle, meaning:

$$\begin{aligned} |QT|^2 =& |QS|^2 + |ST|^2 \\ =& |QF|^2 + |ST|^2 \\ \Rightarrow |QT| >& |QF| \end{aligned}$$

Hence the general point Q isn't on the perpendicular bisector of FT, but P is, hence the perpendicular bisector of FT is the tangent at P.

Analytical

Without loss of generality consider the unit parabola given by:

$$u : y = x^2$$

I state, without proof, that the focus is at $F = \left(0, \frac{1}{4}\right)$ and the directrix is given by:

$$d : y = -\frac{1}{4}$$

The tangent of u at $P = (p, p^2)$ is given by:

$$t : y = 2px - p^2$$

Let $T = \left(p, -\frac{1}{4}\right)$ be the point of directrix such that PT is perpendicular to it. The line connecting the focus to T is given by:

$$a : y = -\frac{x}{2p} + \frac{1}{4}$$

a and t are perpendicular since the product of their gradients is -1:

$$-\frac{1}{2p} \times 2p = -1$$

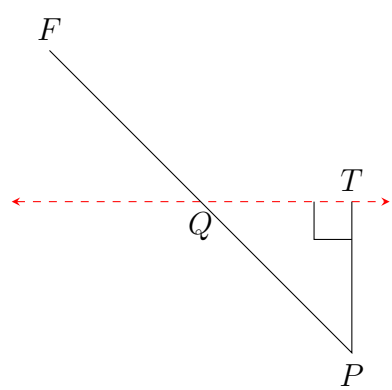
Also notice that the midpoint of FT is given by:

$$M = \left(\frac{1}{2}(p+0), \frac{1}{2}\left(-\frac{1}{4} + \frac{1}{4}\right)\right) = \left(\frac{p}{2}, 0\right)$$

Is a point on both a and t.

2.25.2 Elementary Results

All points on a parabola are on the same side of the directrix as the focus:

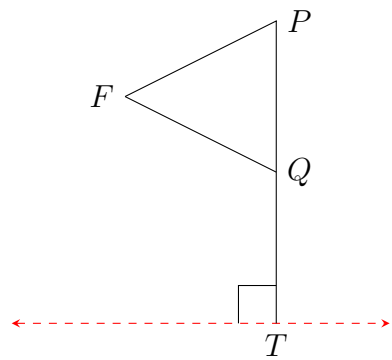


Let P be a point on the parabola on the opposite side of the focus F. Let T be a point on the directrix such that PT is perpendicular to it. Since P and F are on opposite sides of the directrix there is a point Q where the segment FP intersects the directrix.

$$|FP| > |QP| > |TP|$$

Which contradicts $|FP| = |TP|$, hence P doesn't exist.

A line perpendicular to the directrix intersects a parabola at most once:



Let T be the point where a perpendicular line to the directrix intersects the directrix. Let P and Q be points on the same perpendicular line. From the previous results both P and Q are on the same side as the focus F. Without loss of generality let P be the point further away from the directrix.

Since P and Q are points on the parabola we have:

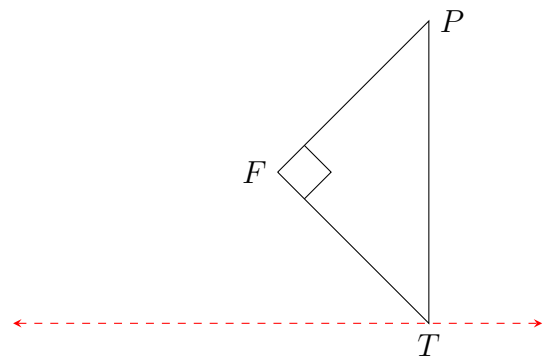
$$|FP| = |PT| \text{ and } |FQ| = |QT|$$

Substituting into the triangle inequality gives:

$$\begin{aligned} |PT| =& |PQ| + |QT| \\ =& |PQ| + |QF| \\ >& |PF| \end{aligned}$$

Which contradicts $|FP| = |TP|$, hence distinct P and Q don't exist.

A line perpendicular to the directrix intersects a parabola at least once:



Let F be the focus and let T be a point on the directrix. Extend a perpendicular line out from T, if the perpendicular intersects F then the midpoint M of FT is on the parabola since:

$$|MF| = |MT|$$

If the perpendicular line doesn't go through F suspend a second perpendicular line from the first perpendicular line at F and let the two lines intersect at the point P.

Let X be a point on the segment PT and define the function:

$$f(X) = |XF| - |XT|$$

We have:

$$f(T) = |TF| - |TT| > 0$$

and since $\angle PFT$ is a right angle we have $|PF| < |PT|$:

$$f(P) = |PF| - |PT| < |PT| - |PT| = 0$$

Hence by continuity there is a point X_0 in the segment PT such that:

$$f(X_0) = 0$$

Meaning:

$$|X_0F| = |X_0T|$$

As required.

A line perpendicular to the directrix intersects a parabola exactly once:

Combine the last two results.

Remarks

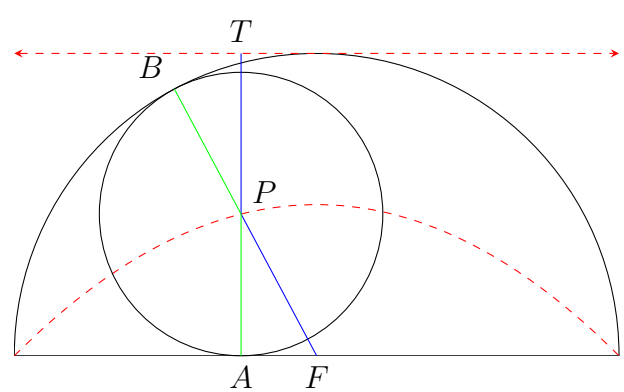
These arguments are non-euclidean in two senses. Neither sense as cool as "non-euclidean" sounds, both of them still interesting.

Firstly, is the use of continuity in the penultimate result. The euclidean axioms famously don't include a concept of continuity. Adding a continuity axiom to Euclid and when the concept is implicitly used has been an ongoing topic of research since the 19th century.

Secondly, is the reverse. Instead of using an another one, we might not need to use all the euclidean axioms. The first results don't use right-angles beyond showing $|PQ| > |PT|$ and the second result doesn't use them at all. To me, this suggests that these results can be generalized to some subset of euclidean geometry. I wasn't able to do so here, but thoughts like this are why I wanted to think about parabolas more.

2.25.3 Construction

Semi-Circle



Take a semicircle with an interior circle such that the interior circle is tangent to both the semicircle arc and base line. Then the centers of the interior

circle forms a parabola.

Let F be the center of the semicircle and P be the center of interior circle. Construct a parallel line to the base line at the apex of arc. Let TA be the perpendicular segment that goes through P and let FB be the radius through P.

By construction $|TA|$ is equal to the radius:

$$|TA| = |FB|$$

Expanding gives:

$$|TP| + |PA| = |FP| + |PB|$$

Since PA and PB are radii on the same interior circle they are of equal length, giving:

$$|TP| = |FP|$$

As required.

2.25.4 Conic Section

Yea it's analytic but synthetic conic section arguments are verbose and I want to note this down somewhere.

Let the cone be given by:

$$x^2 + y^2 - z^2 = 0$$

And the plane by:

$$ay + b - z = 0$$

(Note these coordinates are in the 3D space. If you want the in-plane ones discard the z and scale y by $\sqrt{1+a^2}$).

If $a^2 = 1$ we get:

$$x^2 \pm 2by - b^2 = 0$$

If $a^2 \neq 1$ we get:

$$x^2 + (1-a^2)\left(y - \frac{ab}{1-a^2}\right)^2 - b\,\frac{1+a^2}{1-a^2} = 0$$

2.26 Square Root Limit

While working on something else and the Stack Overflow and the "Hot Network Questions" brought up the following limit:

$$\lim_{x\to\infty}\left(\sqrt{x^2+x+1}-x\right)=\frac{1}{2}$$

The questioner asked why the limit wasn't 0 as $\sqrt{x^2+o(x^2)}\to x$. The simple answer is that you can't split up a sum in a limit like that when each term isn't finite.
An interesting point in this presentation is that x^2+x+1 is manipulated three different ways, if you include not manipulating it as a way, and each manipulation has a different analysis step. Pretty cool to see the algebra-analysis relationship like that.

Algebra Answer

This answer uses the algebra observation:

$$\sqrt{a^2+b}-a=\frac{(\sqrt{a^2+b})^2-a^2}{\sqrt{a^2+b}+a}=\frac{b}{\sqrt{a^2+b}+a}$$

Manipulating the limit into this form gives:

$$\begin{aligned}\sqrt{x^2+x+1}-x=&\sqrt{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}-\left(x+\frac{1}{2}\right)+\frac{1}{2}\\=&\frac{3}{4}\frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}+\left(x+\frac{1}{2}\right)}+\frac{1}{2}\end{aligned}$$

Observe that the denominator grows without bound but the numerator is constant. Hence the fraction goes to 0 meaning only $\frac{1}{2}$ remains, as expected.

L'Hôpital's Rule Answer

This answer isn't as careful with the early algebra and gets less cancellation. Instead it uses L'Hôpital's Rule in a kind of cool way. First perform the

algebra:

$$\begin{aligned}\sqrt{x^2+x+1}-x =& \frac{x^2+x+1-x^2}{\sqrt{x^2+x+1}+x}\\ =& \frac{x+1}{\sqrt{x^2+x+1}+x}\end{aligned}$$

Both numerator and denominator grow without bound, so applying L'Hôpital's rule gives:

$$\sqrt{x^2+x+1}-x \to \left(\frac{2x+1}{2\sqrt{x^2+x+1}}+1\right)^{-1} = (f(x)+1)^{-1}$$

Where:

$$f(x) = \frac{2x+1}{2\sqrt{x^2+x+1}}$$

Similarly, both numerator and denominator grow without bound, so applying L'Hôpital's rule gives:

$$f(x) \to \frac{2}{2\frac{2x+1}{2\sqrt{x^2+x+1}}} = \frac{1}{f(x)}$$

Tidying up gives:

$$\lim_{x\to\infty} f(x) = 1$$

Hence

$$\sqrt{x^2+x+1}-x \to= (f(x)+1)^{-1} = \frac{1}{2}$$

As expected.

Holomorphic Answer

Let a satisfy:

$$a^2 - a + 1 = 0$$

Then let us write:

$$\sqrt{x^2+x+1}-x = \sqrt{(x+a)^2+(1-2a)(x+a)}-(x+a)+a$$

Unfortunately $a \notin \mathbb{R}$ meaning we are now using complex analysis. [9]
Define:

$$h = \frac{1-2a}{x+a}$$

[9] From $(a+1)(a^2-a+1) = a^3+1 = 0$ we know a is an odd 6th root of unity but not -1

Simple algebra gives:

$$\sqrt{x^2+x+1}-x=(1-2a)\frac{\sqrt{1+h}-1}{h}+a$$

Observe that as $|h|\to 0$ we've constructed the derivative of $\sqrt{\cdot}$ at 1 giving:

$$\lim_{|h|\to 0}\left((1-2a)\frac{\sqrt{1+h}-1}{h}+a\right)=(1-2a)\frac{1}{2\sqrt{1}}+a=\frac{1}{2}$$

As expected.

Geometric Progression

If we want the general n^{th}-root consider the geometric progression:

$$(a-b)\sum_{k=0}^{n}a^kb^{n-k}=a^{n+1}-b^{n+1}$$

The difference of two squares we used earlier is the $n=1$ case.

$$\frac{(x+h)^{\frac{1}{n}}-x^{\frac{1}{n}}}{h}=\frac{1}{\sum_{k=0}^{n-1}(x+h)^{\frac{k}{n}}x^{\frac{n-1-k}{n}}}$$

Both numerator and denominator are finite in the $h\to 0$ limit, giving:

$$\lim_{h\to 0}\frac{(x+h)^{\frac{1}{n}}-x^{\frac{1}{n}}}{h}=\frac{1}{nx^{\frac{n-1}{n}}}$$

As expected.

We might be apple to apply a similar argument to a more general problem. Let f,g be functions such the limits $f-g$ and f^p-g^p are finite then we get a cool limit:

$$\sum_{k=0}^{p-1}f^kg^{p-1-k}$$

This is cool since the limit of f^p-g^p looks like it could be related to a cool metric.

Similarly if $f-g$ is finite but both f^p-g^p and $\sum_{k=0}^{p-1}f^kg^{p-1-k}$ are not we could apply L'Hôpital's rule to get an interesting result. The general case has too much algebra for me to care about now, but I might come back to it.